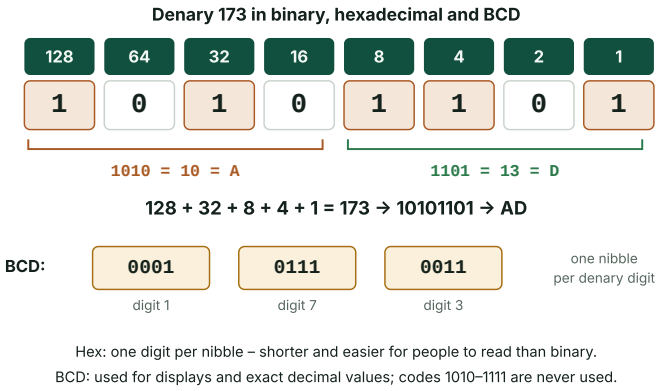
Number systems: bits, bytes, binary, denary, hexadecimal and BCD

FD01 ANSWERS

I can: Define the units of storage from bit to terabyte and convert between them using powers of 2 · Work out how many values n bits can represent and the minimum number of bits needed for a range · Convert confidently between binary, denary and hexadecimal, showing every step · Represent a denary number in BCD and explain where and why BCD is used · Explain why technicians use hexadecimal instead of binary



Use this diagram to help you: 173 converted to binary using place values, split into nibbles to give hexadecimal AD, and stored digit by digit in binary-coded decimal

A Key terms

Write the letter of the matching definition · 6 marks

D **Bit**	**A** A group of 4 bits; half a byte, represented by one hexadecimal digit.
A **Nibble**	**B** 1024 (2^{10}) bytes.
F **Byte**	**C** 1024 kilobytes, or 2^{20} (1 048 576) bytes.
B **Kilobyte (KB)**	**D** A binary digit; the smallest unit of data, which stores either 0 or 1.
C **Megabyte (MB)**	**E** 1024 megabytes, or 2^{30} bytes.
E **Gigabyte (GB)**	**F** A group of 8 bits, which can represent 256 different values.

B Fill the gaps

Use the word bank · 9 marks

1024 · clock · most significant bit · 16 · nibble · bit · 255 · byte · binary-coded decimal

A single **bit** stores a 0 or a 1, a group of four bits is a **nibble** and eight bits make a **byte**. A kilobyte is **1024** bytes. With n bits, 2^n different values can be represented, so the largest unsigned value in one byte is **255**. Hexadecimal is base **16**, and each hex digit maps exactly onto one nibble. In **binary-coded decimal** each denary digit is stored separately in four bits, which is why it is used in a digital **clock**. The leftmost bit, with the highest place value, is the **most significant bit**.

C True or false?

Tick one box · 9 marks

	True	False
One hexadecimal digit represents exactly four bits.	☑	☐
Hexadecimal is used because it takes up less memory than binary. *(The computer stores everything in binary; hex is only a shorter way for people to write the same bits, so it saves no memory.)*	☐	☑
A megabyte is 1024 kilobytes, or 2^{20} bytes.	☑	☐
With 10 bits, 1024 different values can be represented.	☑	☐
In BCD, the nibble 1100 is a valid code. *(BCD only uses 0000 to 1001 for the digits 0–9; 1010 to 1111 are invalid.)*	☐	☑
Storing 99 in BCD needs fewer bits than storing it in pure binary. *(BCD needs 8 bits (1001 1001); pure binary needs only 7 (1100011).)*	☐	☑
In binary, every odd number ends in 1.	☑	☐
To store the values 0 to 256 you need 8 bits. *(0 to 256 is 257 different values, but 8 bits give only 256, so 9 bits are needed.)*	☐	☑
The hex value FF is equal to 255 in denary.	☑	☐

D Exam-style questions

26 marks

1 State how many different values can be represented using 6 bits. [1]

> 2^6 = 64 (1)

2 A sports club in Omagh stores membership numbers from 0 to 1000 as unsigned integers. State the minimum number of bits needed for each membership number and justify your answer. [2]

> 10 bits (1) / 0 to 1000 is 1001 values; 2^9 = 512 is not enough but 2^{10} = 1024 is (1)

3 Convert the 8-bit binary number 10011101 to denary. Show your working. [2]

> Place values / weightings shown: 128 + 16 + 8 + 4 + 1 (1) / = 157 (1)

4 Convert the denary number 201 to 8-bit binary. Show your working. [2]

> Working, e.g. 201 − 128 = 73, 73 − 64 = 9, 9 − 8 = 1, 1 − 1 = 0, or repeated division by 2 with remainders 1,0,0,1,0,0,1,1 (1) / 11001001 (1)

5 Convert the hexadecimal value B7 into (a) binary and (b) denary. Show your working. [3]

> (a) B = 1011, 7 = 0111 (1) / 10110111 (1); (b) 11 × 16 + 7 = 183 (1)

6 Convert the denary number 94 to hexadecimal. Show your working. [2]

> 94 ÷ 16 = 5 remainder 14 (1) / 14 = E so 5E (1)

7 A school technician has a 4 GB memory stick and wants to store backups that are each 512 MB. Calculate how many backups will fit, showing your working. [2]

> 4 GB = 4 × 1024 = 4096 MB (1) / 4096 ÷ 512 = 8 backups (1)

8 A clock on a railway platform in Portadown stores the time using binary-coded decimal. Show how the time 23:58 is stored. [2]

> Each digit coded separately in 4 bits (1) / 0010 0011 0101 1000 (1)

9 Explain two reasons why network technicians at an internet service provider use hexadecimal rather than binary when reading MAC addresses. [4]

> Any two explained points, two marks each: hex is much shorter than binary (1) so it is quicker for technicians to read and write (1) / fewer errors are made when copying or typing values (1) because there are fewer digits to transcribe (1) / easy to convert to binary (1) because each hex digit maps to exactly one nibble (1) / easier to remember or compare addresses (1) with a reason (1). Do not credit 'uses less memory'.

10 A district council is buying car-park pay machines that display charges and record the money taken. Discuss whether amounts of money should be stored in binary-coded decimal or in pure binary. [6]

> Indicative content: BCD codes each denary digit in 4 bits, so pence and pounds can be stored exactly; no rounding error from binary fractions such as 0.10; easy conversion to drive the price display digit by digit; matches how the public and auditors read money. Pure binary needs fewer bits for the same value, so storage is more efficient; arithmetic is simpler and faster in standard hardware; BCD needs correction steps after addition; an alternative is to store money as an integer number of pence in pure binary, which is also exact. Conclusion should be justified, e.g. integer pence or BCD both avoid rounding; BCD favoured for display-driven machines. Levels: 5–6 a balanced discussion of both representations with a justified conclusion and correct technical terms; 3–4 advantages and disadvantages described with limited balance or no conclusion; 1–2 isolated points with little explanation.

E Quick quiz

Circle one answer · 9 marks

1 How many different values can be represented with 8 bits? [1]

A 255 — **B 256 ✓**

C 128 — D 512

2 What is the hexadecimal value 3F in denary? [1]

A 48 — B 36

C 315 — **D 63 ✓**

3 What is the binary number 11010010 in hexadecimal? [1]

A D2 ✓ — B C2

C 2D — D B2

4 Which is the BCD representation of 82? [1]

A 01010010 — B 1000 0001

C 1000 0010 ✓ — D 0101 0010

5 Which of these is NOT a common use of hexadecimal? [1]

A HTML colour codes — B MAC addresses

C Memory addresses in error dumps — **D Storing text in RAM ✓**

6 What is the minimum number of bits needed to store 300 as an unsigned integer? [1]

A 8 — **B 9 ✓**

C 10 — D 16

7 Using the definition credited by CCEA, how many bytes are in a kilobyte? [1]

A 1024 ✓ — B 1000

C 8 — D 1 048 576

8 A file is 2 MB. How many kilobytes is that? [1]

A 2000 — **B 2048 ✓**

C 1024 — D 16

9 What is 214 in 8-bit binary? [1]

A 11010110 ✓ — B 11010101

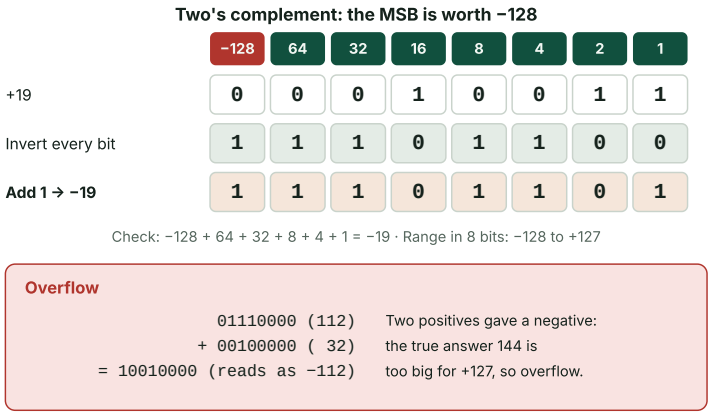
C 10110110 D 11100110

AS 2: FUNDAMENTALS OF DIGITAL TECHNOLOGY · DATA REPRESENTATION

Negative numbers and binary arithmetic

FD02 ANSWERS

I can: Add binary numbers using the four addition rules and show the carries · Represent negative integers in sign and magnitude and in two's complement, and state the range of each · Convert between denary and 8-bit two's complement, checking each answer · Subtract by adding the two's complement and explain why the final carry is discarded · Identify overflow in unsigned and signed arithmetic and explain its consequences



Use this diagram to help you: Forming -19 in 8-bit two's complement by inverting the bits of +19 and adding 1, and an example of overflow when two positive numbers give a negative result

A Key terms

Write the letter of the matching definition · 6 marks

B Binary addition

A Carry

F Sign bit

D Sign and magnitude

C Two's complement

E Unsigned integer

A A bit passed to the next column on the left when a column total is two or more.

B Adding binary numbers column by column from the right using the rules 0+0=0, 0+1=1, 1+1=0 carry 1, 1+1+1=1 carry 1.

C A signed-integer representation in which the MSB has a negative place value; a number is negated by inverting its bits and adding 1.

D A signed-integer representation in which the MSB indicates the sign (0 positive, 1 negative) and the remaining bits give the size.

E A whole number stored with no sign bit, so all bits represent magnitude and only zero or positive values are possible.

F The most significant bit of a signed number, used to show whether the value is positive or negative.

B Fill the gaps

Use the word bank · 9 marks

zero · two's complement · add 1 · carry · status register · negative · sign bit · overflow · adding

Binary addition follows the rule 1 + 1 = 0 **carry** 1. In sign and magnitude the most significant bit is a **sign bit** and the remaining bits give the size, but there are two representations of **zero**. In **two's complement** the MSB has a **negative** place value of −128 in eight bits. To make a number negative, invert the bits and **add 1**. Subtraction is carried out by **adding**

the two's complement of the second number. If two positive numbers are added and the result is negative, **overflow** has occurred, and the processor sets a flag in the **status register**.

C True or false?

Tick one box · 9 marks

	True	False
In 8-bit two's complement, 10000000 represents −128.	☑	☐
Sign and magnitude can represent −128 in 8 bits. *(Its 8-bit range is −127 to +127 because one bit is used only for the sign.)*	☐	☑
In two's complement subtraction, a carry out of the MSB always means overflow. *(When a positive and a negative are added, the carry out is discarded and the result is still correct; overflow is detected from the signs.)*	☐	☑
Two's complement allows a CPU to subtract using its adder circuit.	☑	☐
11111111 in two's complement is −1.	☑	☐
Overflow can only happen with negative numbers. *(Adding two large positive numbers, such as 100 + 50 in 8-bit two's complement, also overflows.)*	☐	☑
In binary, 1 + 1 + 1 gives 1 carry 1.	☑	☐
To find the two's complement of a number you invert the bits and subtract 1. *(You invert the bits and then add 1.)*	☐	☑
Adding a positive number to a negative number in two's complement can never cause overflow.	☑	☐

D Exam-style questions

30 marks

1 State the range of integers that can be represented in 8-bit two's complement. [1]

> −128 to +127 (1)

2 Represent −45 in 8-bit sign and magnitude. [1]

> 10101101 (1)

3 Represent −45 in 8-bit two's complement. Show your working. [2]

> 45 = 00101101, invert to 11010010 (1) / add 1 = 11010011 (1)

4 Convert the 8-bit two's complement number 11101100 to denary. Show your working. [2]

> −128 + 64 + 32 + 8 + 4 shown (1) / = −20 (1)

5 Add the unsigned binary numbers 01101110 and 00101011. Show your working, including carries, and give the answer in binary and denary. [3]

> Correct addition with carries shown (1) / 10011001 (1) / = 153 (1)

6 A stock system in a Derry hardware shop stores changes in stock as 8-bit two's complement integers. Using two's complement, calculate 38 − 52. Show your working and give the answer in denary. [4]

> 38 = 00100110 and 52 = 00110100 (1) / two's complement of 52 = 11001100 (1) / 00100110 + 11001100 = 11110010 (1) / = −128 + 64 + 32 + 16 + 2 = −14 (1)

7 Explain two problems with using sign and magnitude to represent negative numbers in a processor. [4]

Two zeros exist, +0 and −0 (1) wasting a pattern / making comparisons with zero more complex (1) / Ordinary binary addition gives wrong answers when signs differ (1) so extra circuitry is needed to compare signs and decide whether to add or subtract (1) / Accept: range is one smaller than two's complement (1) because a pattern is lost to −0 (1)

8 A hospital heart-rate monitor stores readings as 8-bit unsigned integers. Explain what is meant by overflow, using the addition 180 + 90 as an example. [3]

Overflow: result too large to be stored in the available bits (1) / 180 + 90 = 270, which is greater than 255 (1) / the carry out of the MSB is lost, so the stored result is 00001110 = 14, which is wrong (1)

9 Explain how overflow can be detected when adding two 8-bit two's complement numbers, and explain why a carry out of the most significant bit does not always mean the answer is wrong. [4]

Overflow if two numbers with the same sign are added (1) and the result has the opposite sign (1) / when a positive and a negative are added the result always lies within range (1) so the carry out of the MSB is simply discarded and the 8-bit result is correct, e.g. 45 + (−19) = 26 (1)

10 A team is writing software for a cold-store temperature logger at a Co. Tyrone food company. Readings range from −30 °C to +40 °C and differences between readings are calculated. Evaluate the choice of 8-bit sign and magnitude, 8-bit two's complement or a larger data type for storing these values. [6]

Indicative content: range −30 to +40 fits all three options. Sign and magnitude: range −127 to +127 is enough, easy to read, but two zeros complicate comparisons with 0 °C and arithmetic on differences needs extra logic. Two's complement: range −128 to +127 is enough, one zero, differences calculated with ordinary addition; but differences between readings can be up to 70, and sums or totals of many readings could overflow 8 bits. Larger type (e.g. 16-bit or 32-bit int): no overflow risk for totals or averages, but uses more memory per reading; may matter for a logger storing very many readings; whole-degree integers lose precision, so a fixed-point or real type may be considered. Justified conclusion, e.g. two's complement in a larger integer type. Levels: 5–6 evaluates all options with technical accuracy and a justified conclusion; 3–4 describes advantages and disadvantages of at least two options; 1–2 basic points with little justification.

E Quick quiz

Circle one answer · 9 marks

1 What denary value does 11110110 represent in 8-bit two's complement? [1]

A −10 ✓ B 246
C −118 D −9

2 Why is two's complement preferred to sign and magnitude in processors? [1]

A It is easier for people to read B It uses fewer bits for each number
C Subtraction can be done using the same adder circuit ✓ D It can store real numbers

3 What is the range of 8-bit sign and magnitude? [1]

A −128 to +127 B 0 to 255
C −128 to +128 **D −127 to +127 ✓**

4 Which addition overflows in 8-bit two's complement? [1]

A 00010000 + 00010000 **B 01100000 + 00110000 ✓**
C 11110000 + 00000100 D 10000000 + 01111111

5 What is 1 + 1 + 1 in binary addition? [1]

A 0 carry 1 B 1 carry 0
C 10 carry 1 **D 1 carry 1 ✓**

6 What is −82 in 8-bit two's complement? [1]

A 11010010 **B 10101110 ✓**
C 10101101 D 01010010

7 In the subtraction 45 − 19 using 8-bit two's complement, what happens to the ninth bit produced? [1]

A It is discarded ✓
B It shows overflow has occurred
C It becomes the sign bit
D It is added back to the LSB

8 What is the unsigned 8-bit result of 200 + 100? [1]

A 300
B 00101100 (44) because of overflow ✓
C 11111111 (255)
D An error with no stored result

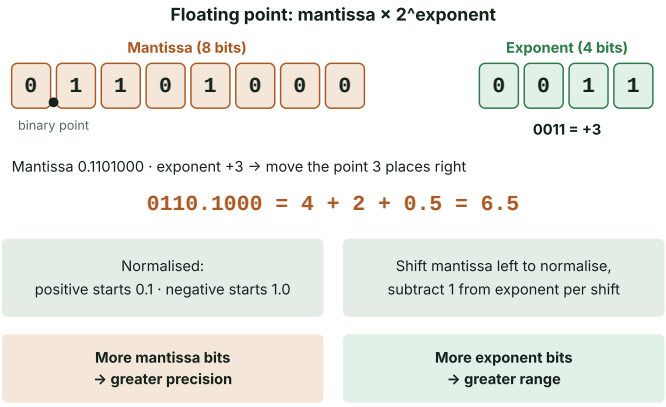
9 How many different values can 8-bit two's complement represent? [1]

A 255
B 127
C 256 ✓
D 128

Real numbers: fixed-point and floating-point

FD03 ANSWERS

I can: Convert between denary and fixed-point binary with a given binary point position · Convert floating-point numbers (two's complement mantissa and exponent) to denary and back · Normalise a floating-point number and explain why normalisation is used · Explain how the split of bits between mantissa and exponent affects range and precision · Explain why some real numbers cannot be stored exactly and the effect this has on programs



Use this diagram to help you: A floating-point number with an 8-bit two's complement mantissa and 4-bit exponent; the exponent of 3 moves the binary point three places right to give 6.5

A Key terms

Write the letter of the matching definition · 6 marks

B Real number

D Fixed-point representation

F Binary point

E Floating-point representation

C Mantissa

A Exponent

A The part of a floating-point number that states how many places, and in which direction, the binary point is moved.

B A number that can have a fractional part, such as 6.75 or −0.3125.

C The part of a floating-point number that holds its significant digits.

D A way of storing real numbers in which the binary point is at an agreed, unchanging position.

E A way of storing real numbers as a mantissa multiplied by 2 to the power of an exponent, allowing a wide range of values.

F The point separating the whole-number bits from the fractional bits, whose place values are ½, ¼, ⅛ and so on.

B Fill the gaps

Use the word bank · 8 marks

rounding error | binary point | exponent | precision | mantissa | underflow | range | normalised

In fixed-point binary the **binary point** is in an agreed position, so precision is the same across the whole range. A floating-point number is made up of a **mantissa** and an **exponent**. Increasing the number of bits in the exponent increases the **range** of values, while increasing the bits in the mantissa increases **precision**. A positive **normalised** mantissa begins with 0.1 and a negative one

begins with 1.0. The value 0.1 cannot be stored exactly in binary, which causes a **rounding error**. A number that is too close to zero to be stored causes **underflow** .

C True or false?

Tick one box · 9 marks

	True	False
A normalised negative mantissa begins with 1.0.	☑	☐
Increasing the bits in the mantissa increases the range of numbers. *(More mantissa bits increase precision; more exponent bits increase range.)*	☐	☑
The denary value 0.1 can be stored exactly in binary floating point. *(0.1 recurs in binary (0.000110011…), so it is stored as an approximation.)*	☐	☑
Fixed-point arithmetic is generally faster than floating-point arithmetic.	☑	☐
Normalisation gives each number a single representation.	☑	☐
When normalising, the exponent is increased by one for each left shift. *(Shifting the mantissa left multiplies it by 2, so the exponent is decreased by one to keep the value the same.)*	☐	☑
A negative exponent moves the binary point to the left.	☑	☐
Underflow means a result is a negative number that is too large to store. *(Underflow means the value is too close to zero to represent, so it becomes zero.)*	☐	☑
In 4.4 unsigned fixed point, the largest value is 15.9375.	☑	☐

D Exam-style questions

30 marks

1 State what is meant by the mantissa of a floating-point number. [1]

> The part of a floating-point number that holds the significant digits/bits of the value (1)

2 Convert the fixed-point number 0101.1100 (4 bits before and 4 bits after the binary point) to denary. Show your working. [2]

> 4 + 1 + 0.5 + 0.25 shown (1) / = 5.75 (1)

3 Represent 3.625 in unsigned fixed-point binary using 4 bits before and 4 bits after the binary point. [2]

> 3 = 0011 (1) / .625 = .1010, giving 0011.1010 (1)

4 A floating-point number uses an 8-bit two's complement mantissa followed by a 4-bit two's complement exponent: 0.1001000 0101. Convert it to denary, showing your working. [3]

> Exponent 0101 = +5 (1) / point moved 5 places right: 010010.00 (1) / = 16 + 2 = 18 (1)

5 Using the same format, convert 1.0110000 1111 to denary, showing your working. [3]

> Exponent 1111 = −1 (1) / mantissa 1.0110000 = −1 + 0.25 + 0.125 = −0.625, or point moved 1 place left with sign extension giving 1.10110000 (1) / $-0.625 \times 2^{-1} = -1 + 0.5 + 0.125 + 0.0625 = -0.3125$ (1)

6 Normalise the floating-point number 0.0001100 0100 (8-bit mantissa, 4-bit exponent). Show the new mantissa and exponent. [2]

> Mantissa 0.1100000 (1) / exponent 0001 (1)

7 Represent −3 as a normalised floating-point number using an 8-bit two's complement mantissa and a 4-bit two's complement exponent. Show your working. [3]

+3 = 11 = 0.1100000 × 2^2 (1) / two's complement of mantissa: 1.0100000 (1) / exponent 0010, so 1.0100000 0010 (1)

8 Explain why floating-point numbers are normalised. [2]

Maximises precision by using all mantissa bits for significant digits / removing redundant leading bits (1) / gives each value a unique representation, making comparison easier (1)

9 A data logger at a Co. Antrim weather station stores readings in 16-bit floating point. Explain the effect of changing from a 10-bit mantissa and 6-bit exponent to a 12-bit mantissa and 4-bit exponent. [4]

More mantissa bits (10 → 12) increase precision (1) because more significant bits are stored (1) / fewer exponent bits (6 → 4) reduce the range (1) so very large or very small readings may not be storable / may overflow or underflow (1)

10 A software house in Belfast is writing a payroll system for a health trust and a physics simulation for a university. Discuss the use of fixed-point, floating-point and other representations of real numbers in these two systems. [8]

Indicative content. Payroll: money must be exact; floating point causes rounding errors (0.1 not exact) which accumulate over many payments; use integer pence, fixed point or a decimal/BCD type; range of salaries is known and modest, so fixed point is sufficient and fast; legal/audit need for exact totals. Physics simulation: needs a very wide range (very large and very small quantities), so floating point is appropriate; relative precision acceptable; double gives about 15–16 significant digits; normalisation maximises precision; risk of overflow/underflow and accumulated rounding error must be managed, e.g. compare with tolerances; floating-point hardware makes it fast enough. Comparison of range versus precision and speed. Levels: 7–8 thorough discussion of both systems with accurate terminology and justified choices; 4–6 sound discussion of both with some justification or one system in depth; 1–3 limited points, mostly descriptive.

E Quick quiz

Circle one answer · 9 marks

1 In floating point, what is the main effect of using more bits for the exponent and fewer for the mantissa? [1]

A Greater precision
B Faster processing
C Greater range but less precision ✓
D Fewer rounding errors

2 Why are floating-point numbers normalised? [1]

A To make them positive
B To maximise precision and give a unique representation ✓
C To increase the range of the exponent
D To convert them to fixed point

3 What is 0.1100000 0010 (8-bit mantissa, 4-bit exponent) in denary? [1]

A 0.75
B 1.5
C 6
D 3 ✓

4 Which mantissa is normalised? [1]

A 0.0110000
B 1.1010000
C 0.1010000 ✓
D 1.1100000

5 What does 0110.0100 represent in unsigned 4.4 fixed point? [1]

A 6.25 ✓
B 6.4
C 6.5
D 100.25

6 What is underflow? [1]

A A number too large to store
B A carry out of the MSB
C A rounding error in BCD
D A number too close to zero to represent ✓

7 What is 0.1100000 1110 in denary? [1]

A 0.1875 ✓ B 0.75

C 3 D −0.25

8 Normalise 0.0010110 0110. What is the new exponent? [1]

A 1000 **B 0100 ✓**

C 0101 D 0110

9 Why is money often stored as an integer number of pence? [1]

A It uses less memory than any other type **B Values such as £0.10 cannot be stored exactly as binary fractions ✓**

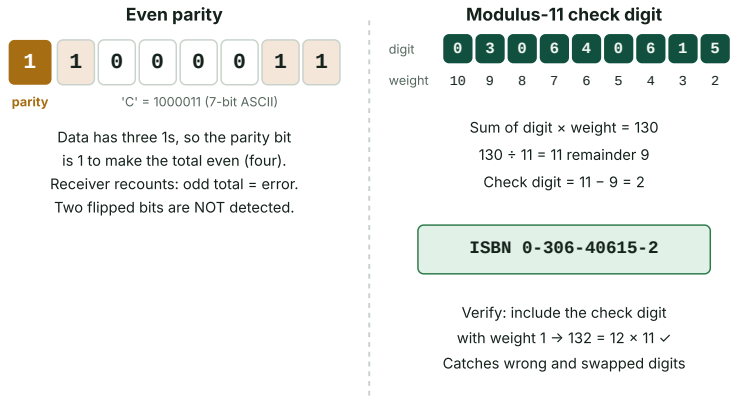
C Integers cannot overflow D Floating point cannot store negative numbers

AS 2: FUNDAMENTALS OF DIGITAL TECHNOLOGY · DATA REPRESENTATION

Character sets and error detection

FD04 ANSWERS

I can: Explain how a character set maps each character to a unique binary code, and use the ordering of ASCII codes to work out unknown codes · Compare ASCII, extended ASCII and Unicode (UTF-8, UTF-16, UTF-32) in terms of range of characters and bytes needed · Add and check odd or even parity bits and explain why a single parity bit cannot catch every error · Explain how a check digit catches keying errors such as transpositions, and how this differs from parity checking of transmitted data



Use this diagram to help you: An even parity bit added to the ASCII code for C, and a modulus-11 check digit calculation for an ISBN-10

A Key terms

Write the letter of the matching definition · 6 marks

B Character set

C ASCII

F Extended ASCII

E Unicode

D UTF-8

A Control character

A A non-printing character code, such as carriage return, line feed or tab, used to control how text is processed or displayed.

B An agreed list of characters that a computer system can represent, in which each character is given a unique binary code.

C American Standard Code for Information Interchange: a 7-bit character set of 128 codes covering English letters, digits, punctuation and control characters.

D A variable-length Unicode encoding that uses one byte for ASCII characters and up to four bytes for other characters.

E A character standard that gives a unique code point to characters from almost all of the world's writing systems, plus symbols and emoji; its first 128 codes match ASCII.

F An 8-bit extension of ASCII giving 256 codes; codes 0–127 match ASCII and codes 128–255 hold accented letters and symbols, which vary between code pages.

B Fill the gaps

Use the word bank · 8 marks

transposition error | parity bit | Unicode | check digit | character set | UTF-8 | ASCII | even parity

A **character set** is an agreed list of characters with a unique binary code for each. Standard **ASCII** uses 7 bits, giving 128 codes, while extended ASCII uses 8 bits for 256. **Unicode** gives a code point to characters from almost every language, and its **UTF-8** encoding uses between one and four bytes per character. A **parity bit** is added to each byte so that, with **even parity**, the total number of 1s is even. A **check digit** is calculated from the other digits of a code and catches keying mistakes made at data entry. Swapping two digits is called a **transposition error**.

C True or false?

Tick one box · 9 marks

	True	False
Extended ASCII can represent 256 different characters.	☑	☐
UTF-8 always uses two bytes for every character. *(UTF-8 uses one to four bytes; ASCII characters need just one byte, accented Latin letters two, and most emoji four.)*	☐	☑
In ASCII, the code for a lower-case letter is 32 more than the code for the same capital letter.	☑	☐
A single parity bit shows which bit in a byte has been corrupted. *(A parity bit only shows that the byte contains an error; the receiver cannot tell which bit, so the byte is re-sent.)*	☐	☑
If two bits in the same byte are flipped, even parity will still detect the error. *(Two flips leave the number of 1s even (or odd), so the parity is unchanged and the error is missed.)*	☐	☑
The first 128 Unicode code points are the same as the ASCII codes.	☑	☐
The Modulus 11 check digit method detects transposition errors such as typing 1243 instead of 1234.	☑	☐
A valid check digit proves that the code belongs to the correct customer. *(A check digit only shows the code is a valid one; it could still be another customer's valid number.)*	☐	☑
Codes with modulus-11 check digits are best stored as integers. *(They should be stored as strings, because the check digit can be X and leading zeros must be kept.)*	☐	☑

D Exam-style questions

32 marks

1 State what is meant by a character set. [2]

> An agreed set of characters that a computer system can recognise and represent (1) / each character has its own unique binary code/number (1).

2 The ASCII code for 'K' is 75. State the ASCII code for 'N' and the ASCII code for 'k'. [2]

> 'N' = 78 (1) / 'k' = 107 (1).

3 A pupil stores the digit '5' in a string variable and the number 5 in an integer variable. Explain why these are stored differently. [2]

> The character '5' is stored as its character code (53 in ASCII/Unicode) (1) / the integer 5 is stored as the binary value of 5 (00000101), so arithmetic can only be done directly on the integer (1).

4 A language school in Belfast keeps learners' names and messages in Irish, Polish and Mandarin. Explain why its database should use Unicode rather than ASCII, and state one drawback of doing so. [4]

> ASCII only has 128 (or 256 extended) characters (1) / it cannot represent fadas/Polish accented letters/Mandarin characters (1) / Unicode has a code point for characters in almost all languages so all names and messages display correctly (1). Drawback: characters can take more than one byte (up to 4 in UTF-8), so files/storage/transmission are larger (1).

5 Using even parity, state the parity bit that would be added to the 7-bit code 1101101. [1]

> 1 (five 1s in 1101101, so a 1 is needed to make the total even) (1).

6 A water-quality buoy on Lough Neagh sends readings to the shore as 7-bit ASCII with odd parity. The byte 10011010 is received. State whether an error has been detected and explain your answer. [2]

> Yes, an error has been detected (1) / the byte contains four 1s, which is even, but odd parity requires an odd number of 1s (1).

7 Describe two limitations of using a single parity bit to detect transmission errors. [4]

> Any two, described, from: if an even number of bits (e.g. two) are flipped, the number of 1s is still correct so the error is not detected (2) / it shows that an error has occurred but not which bit is wrong, so the error cannot be corrected and data must be re-sent (2) / extra bit per byte adds overhead/reduces data rate (2) / both devices must agree odd or even in advance (2).

8 A bus stop display in Omagh receives the word 'Bus' as 7-bit ASCII with an even parity bit added on the left of each code. The codes are B = 1000010, u = 1110101 and s = 1110011. Write the three bytes that are sent. [3]

> B has two 1s, so parity bit 0: 01000010 (1) / u has five 1s, so parity bit 1: 11110101 (1) / s has five 1s, so parity bit 1: 11110011 (1).

9 A leisure centre's membership numbers end in a check digit, and its card readers send data to the central server with a parity bit on every byte. Explain why both checks are used, stating which kind of error each one detects. [4]

> Check digit: recalculated when a membership number is keyed in or scanned (1) and detects data-entry errors such as a mistyped digit or two digits swapped/transposition (1). Parity bit: checked by the server for every byte received (1) and detects bits flipped/corrupted during transmission, so the byte can be re-sent (1).

10 A health trust in Northern Ireland sends patient records between hospital sites, and reception staff key in patient reference numbers. Discuss how parity checking and check digits could help keep this data accurate. Your answer should consider what each method can and cannot detect. [8]

> Indicative content: Parity: a parity bit is added to each byte so the count of 1s is odd/even as agreed; receiving site recounts; a mismatch shows corruption in transmission and the byte/block is re-sent; cannot locate the error; misses even numbers of flipped bits; adds overhead; other methods (checksums/CRC) used for whole records. Check digits: calculated from the other digits (e.g. Modulus 11) and stored with the patient number; recalculated when receptionist keys it in; detects single-digit errors and transpositions at input; cannot detect a valid number for the wrong patient, so other checks (e.g. date of birth lookup) still needed. Overall: the two methods protect different stages (input vs transmission) and together reduce but do not eliminate errors. Levels: 7–8 both methods explained accurately with limitations and a reasoned conclusion in context; 4–6 both methods described with some limitations, partly in context; 1–3 limited description of one or both methods, little context.

E Quick quiz

Circle one answer · 9 marks

1 How many different characters can 7-bit ASCII represent? [1]

A 64 **B 128 ✓**

C 256 D 65,536

2 The ASCII code for 'G' is 71. What is the ASCII code for 'g'? [1]

A 71 B 97

C 103 ✓ D 135

3 Which byte is correct if ODD parity is being used? [1]

A 11000011 B 01100110

C 10110101 ✓ D 11110000

4 A byte arrives as 11100101 on a link that uses even parity. What should the receiver do? [1]

A Ask for the byte to be sent again ✓ B Accept the byte as correct

C Flip the parity bit to correct it D Switch to odd parity

5 Why is UTF-8 described as backwards compatible with ASCII? [1]

A It only uses 7 bits B It has no control characters

C It adds a parity bit to each byte **D Its first 128 codes are identical to ASCII and each takes one byte ✓**

6 Which error would a single even parity bit NOT detect? [1]

A Two bits flipped in the same byte ✓ B One data bit flipped

C Three bits flipped in the same byte D The parity bit itself flipped

7 Which error is a check digit designed to catch? [1]

A A bit flipped by interference on a cable B A virus hidden in an email attachment

C Two digits of a membership number typed in the wrong order ✓ D A file saved in the wrong folder

8 How many bytes does the character 'é' take in UTF-8? [1]

A 1 **B 2 ✓**

C 3 D 4

9 What is the main problem with extended ASCII that Unicode solves? [1]

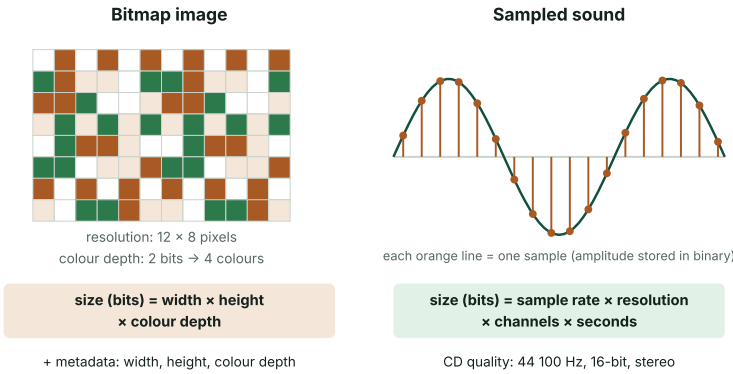
A It cannot store digits B It uses too many bytes per character

C It cannot store control characters **D Codes 128–255 mean different characters on different code pages ✓**

Representing images and sound

FD05 ANSWERS

I can: Explain how pixels, resolution and colour depth determine the quality and file size of a bitmap image · Calculate the uncompressed size of an image from its pixel dimensions or from inches and dpi · Compare bitmap and vector graphics and justify which suits a given use · Explain how an ADC samples a sound wave and calculate uncompressed sound file sizes



Use this diagram to help you: A bitmap as a grid of pixels with the image file-size formula, and a sound wave sampled at regular intervals with the sound file-size formula

A Key terms

Write the letter of the matching definition · 6 marks

B Pixel

C Resolution

F Colour depth

A Metadata

E Bitmap

D Vector graphic

A Data about data, such as an image's width, height and colour depth, stored with the file so that software can interpret it correctly.

B Short for picture element: the smallest addressable element of a bitmap image, storing a single colour as a binary value.

C The number of pixels in an image, usually given as width × height in pixels, or as dots per inch (dpi) for printed images.

D An image stored as a set of objects described by mathematical properties such as coordinates, size and colour, so it can be scaled without loss of quality.

E An image stored as a grid of pixels, with the colour of each pixel held as a binary number.

F The number of bits used to represent the colour of each pixel; n bits give 2^n possible colours, and more bits give a larger file.

B Fill the gaps

Use the word bank · 9 marks

analogue-to-digital converter | pixels | vector | colour depth | sample resolution | metadata | resolution | channels | sampling rate

A bitmap image is made up of a grid of **pixels**. The number of pixels in the image is its **resolution**, and the number of bits used for each pixel is the **colour depth**. Data stored in the file

about the image, such as its width and height, is called **metadata**. A **vector** graphic stores shapes as objects with properties, so it does not pixelate when enlarged. Sound is digitised by an **analogue-to-digital converter**, which takes samples of the wave's amplitude. The number of samples taken per second is the **sampling rate**, and the bits used for each sample is the **sample resolution**. Uncompressed sound file size = sampling rate × sample resolution × number of **channels** × duration in seconds.

C True or false?

Tick one box · 9 marks

	True	False
A colour depth of 8 bits allows 256 different colours.	☑	☐
Enlarging a vector graphic makes it pixelated. *(A vector is redrawn from the mathematical description of its objects, so it stays sharp at any size.)*	☐	☑
Without metadata such as width and height, software could not rebuild a bitmap's pixel grid correctly.	☑	☐
Doubling the sampling rate halves the file size of a recording. *(Doubling the sampling rate doubles the number of samples, so the file size doubles.)*	☐	☑
A stereo recording needs twice the storage of a mono recording made with the same settings.	☑	☐
Vector graphics are the best choice for storing photographs. *(Photographs contain continuous variations of colour, which bitmaps represent far better than shapes can.)*	☐	☑
24-bit colour uses 8 bits each for the red, green and blue components of a pixel.	☑	☐
The sampling rate should be at least twice the highest frequency to be recorded.	☑	☐
The size of a vector file depends on how large the image is displayed. *(Vector file size depends on the number and complexity of the objects, not the display size.)*	☐	☑

D Exam-style questions

30 marks

1 State what is meant by the colour depth of an image. [1]

> The number of bits used to represent the colour of each pixel (1).

2 State how many different colours can be represented with a colour depth of 16 bits. [1]

> 2^{16} (1) = 65,536 colours. [1 mark for 65,536]

3 Calculate the file size, in kilobytes, of an uncompressed 640 × 480 pixel image with a colour depth of 8 bits. Show your working. [3]

> 640 × 480 = 307,200 pixels (1) / × 8 bits = 2,457,600 bits ÷ 8 = 307,200 bytes (1) / ÷ 1024 = 300 KB (1).

4 A school yearbook photo measuring 5 inches by 3 inches is scanned at 300 dpi with 24-bit colour. Calculate its uncompressed size in megabytes. Show your working. [4]

> Width 5 × 300 = 1500 pixels and height 3 × 300 = 900 pixels (1) / 1500 × 900 = 1,350,000 pixels (1) / × 3 bytes (24 bits) = 4,050,000 bytes (1) / ÷ 1024 ÷ 1024 ≈ 3.86 MB (1). [Accept 4.05 MB if 1000 used, with working.]

5 Explain why an image file must store metadata, and state two items of metadata that are needed to display the image correctly. [3]

> Software needs metadata to reconstruct the pixel grid / interpret the bit pattern correctly (1). Any two from: width (1) / height (1) / colour depth (1) / [accept resolution in dpi, file type, compression type, palette].

6 Describe how an analogue sound is converted into digital data. [3]

> A microphone converts the sound wave into an analogue electrical signal (1) / an analogue-to-digital converter (ADC) measures/samples the amplitude at regular intervals (1) / each sample is stored as a binary number using a fixed number of bits (sample resolution) (1).

7 Explain the effect of increasing the sampling rate on a sound recording. [2]

> More samples are taken per second so the digital wave is a closer match to the original/higher frequencies captured/better quality (1) / file size increases in proportion (1).

8 A community radio station in Newry records a 5-minute interview at 48,000 Hz with a 24-bit sample resolution in stereo. Calculate the uncompressed file size in megabytes. Show your working. [3]

> 5 × 60 = 300 s (1) / 48,000 × 24 × 2 × 300 = 691,200,000 bits (1) / ÷ 8 = 86,400,000 bytes ÷ 1024 ÷ 1024 ≈ 82.4 MB (1). [Accept 86.4 MB using 1000.]

9 An Ulster rugby supporters' club records phone messages at a lower sampling rate and sample resolution than CD audio. Explain one advantage and one disadvantage of this choice. [4]

> Advantage: fewer samples and fewer bits per sample give a smaller file (1) / so less storage is needed and messages upload/download faster (1). Disadvantage: the recording is a less accurate match to the original wave (1) / so voices sound less clear / high frequencies are lost / more noise (1).

10 A garden centre in Enniskillen needs a new logo for its website, printed signs, loyalty cards and delivery vans, and also wants photographs of its plants for its online shop. Discuss whether bitmap or vector graphics should be used for each purpose. [6]

> Indicative content: Logo → vector: stored as objects with properties; scales to van-sized or card-sized without pixelation; one small file for all uses; easy to recolour/edit; flat colours suit vector. Photographs → bitmap: continuous colour/tone needs pixels; high colour depth (24-bit) for realistic plants; larger files, so compress (e.g. JPEG) for the online shop to keep load times low; would pixelate if enlarged so capture at suitable resolution. Conclusion: use both, each for its strengths. Levels: 5–6 both uses discussed with accurate technical reasons and justified conclusion; 3–4 reasonable discussion, some technical reasons, may be unbalanced; 1–2 simple statements with little justification.

E Quick quiz

Circle one answer · 9 marks

1 How many colours can be represented with a colour depth of 4 bits? [1]

A 4 — B 8
C 16 ✓ — D 256

2 What is the uncompressed size of a 100 × 100 pixel image with 24-bit colour? [1]

A 30,000 bytes ✓ — B 240,000 bytes
C 10,000 bytes — D 2,400 bytes

3 Which term describes the number of bits used to store each sound sample? [1]

A Sampling rate — **B Sample resolution ✓**
C Number of channels — D Dots per inch

4 Which is the best reason to store a road sign design as a vector graphic? [1]

A It looks more like a photograph — B It records light and shade more accurately
C It always contains GPS metadata — **D It can be printed at any size without losing quality ✓**

5 Which device converts the analogue signal from a microphone into digital samples? [1]

A DAC — B ALU

C ADC ✓ D GPU

6 A 1-minute mono recording is made at 8,000 Hz with 8-bit samples. What is its size? [1]

A 480,000 bytes ✓ B 64,000 bytes

C 3,840,000 bytes D 60,000 bytes

7 Which item of metadata is essential for displaying a bitmap correctly? [1]

A The camera model **B The image width in pixels ✓**

C The GPS location D The date taken

8 Why does CD audio use a sampling rate of 44,100 Hz? [1]

A It is more than twice the highest frequency people can hear ✓ B It gives 65,536 amplitude levels

C It matches the number of pixels on a CD D It is the maximum an ADC can manage

9 A photo is 1200 × 800 pixels with 24-bit colour. What is its size in bytes? [1]

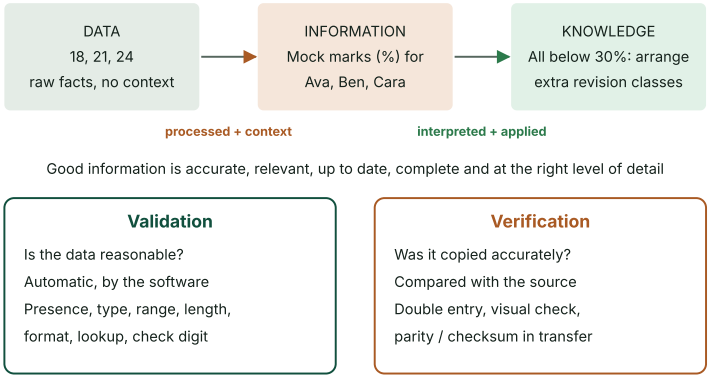
A 960,000 B 23,040,000

C 2,880,000 ✓ D 7,680,000

Data, information and knowledge; quality of information; validation and verification

FD06 ANSWERS

I can: Use one real example to show how raw data becomes information and then knowledge · Evaluate the quality of information using accuracy, relevance, currency, completeness, presentation and reliability · Choose and describe a suitable validation check, with its exact rule, for a given field · Compare validation with verification and explain why neither can guarantee that data is correct



Use this diagram to help you: Data becomes information when it is processed and given context, and information becomes knowledge when it is interpreted and applied; validation and verification compared

A Key terms

Write the letter of the matching definition · 6 marks

D **Data**	A Data that has been processed and given context so that it has meaning to the person using it.
A **Information**	B An input error in which data is copied or typed incorrectly, such as 1500 instead of 150.
F **Knowledge**	C Garbage in, garbage out: if the input data is of poor quality, the output information will also be of poor quality.
C **GIGO**	D Raw facts and figures, such as numbers, text, images or sound, that have no context and therefore no meaning on their own.
E **Transposition error**	E An input error in which two characters are entered in the wrong order, such as 47019 instead of 40719.
B **Transcription error**	F The result of applying one or more rules to information so that an informed decision can be made.

B Fill the gaps

Use the word bank · 10 marks

data · transposition · double entry · verification · validation · knowledge · information · range check · GIGO · lookup check

Raw facts and figures with no context are called **data**. When data is processed and given context it becomes **information**. Applying a rule to information so that a decision can be made produces **knowledge**. Poor input producing poor output is summed up as **GIGO**. Swapping

two digits when typing is a **transposition** error. Information that reflects the current situation is described as up to date. An automatic check that data is sensible and obeys rules is called **validation**; a check that a value lies between two limits is a **range check**, and a check against a list of valid values is a **lookup check**. Typing a new password twice so the computer can compare them is **double entry**, a method of **verification**.

C True or false?

Tick one box · 9 marks

	True	False
Validation guarantees that data is correct. *(Validation only checks that data is sensible and reasonable; a valid value can still be wrong.)*	☐	☑
Double entry is a method of verification.	☑	☐
Knowledge is produced by applying a rule to information so that a decision can be made.	☑	☐
The more detailed information is, the higher its quality for every user. *(Too much detail can hide what matters; information must be relevant and at the right level of detail for its user.)*	☐	☑
A format check on DD/MM/YY would reject the date 45/33/87. *(45/33/87 matches the pattern of two digits, slash, two digits, slash, two digits, so it passes a format check; range checks are also needed.)*	☐	☑
Entering 47019 instead of 40719 is a transposition error.	☑	☐
Proofreading is best done by the same person who typed in the data. *(A different person is more likely to spot mistakes than the person who made them.)*	☐	☑
If a source document contains an error, verification will not detect it.	☑	☐
A presence check makes sure a field has not been left blank.	☑	☐

D Exam-style questions

41 marks

1 A hospital ward records patients' temperatures. Using an example from this ward, show how data, information and knowledge differ. [3]

> Data: a raw value with no context, e.g. 38.9 (1) / Information: the data with context/meaning, e.g. patient in bed 4 had a temperature of 38.9 °C at 14:00 (1) / Knowledge: a rule applied to the information to make a decision, e.g. rule 'above 38 °C means fever', so the nurse alerts the doctor (1)

2 Explain what is meant by 'garbage in, garbage out' (GIGO). [2]

> If the input data is of poor quality / inaccurate (1) / the output information will also be poor / inaccurate, however good the processing system is (1)

3 A pharmacy technician enters a patient number of 40719 as 47019. State the type of error that has been made and explain why validation may not detect it. [3]

> Transposition error (1) / 47019 is still a sensible / reasonable patient number, e.g. correct length, type and format (1) / validation only checks data is reasonable, not that it is correct (1). Credit: a check digit could detect it.

4 A travel agent in Newry gives customers information about holidays. Explain why this information must be relevant and complete if customers are to make good choices. [4]

> Relevance: the information should only include what the customer needs, e.g. only holidays to the requested destination and dates (1) / irrelevant details waste time and can hide useful options (1). Completeness: all the information needed to decide must be present, e.g. total price including transfers, or flight connection times (1) / missing details could lead to a poor or costly choice (1)

5 State the validation check that would reject a mark of 105 for a test marked out of 100. [1]

> Range check (1)

6 A leisure centre in Dungannon uses an online membership form. Name a suitable validation check, and give its rule, for each of the following fields: (a) membership number, two letters followed by five digits; (b) number of children attending, from 0 to 6; (c) email address, which must be supplied; (d) membership type, which must be Adult, Junior or Senior. [8]

> (a) Format check (1): two letters followed by five digits, e.g. LLDDDDD (1) / (b) Range check (1): >= 0 AND <= 6 (1) / (c) Presence check (1): field must not be blank (1) / (d) Lookup check (1): value must be in the list Adult, Junior, Senior (1). Accept length check for (a) with rule 7 characters, and type check with integer for (b) for the first mark only.

7 Compare data validation and data verification. [4]

> Both are used to reduce errors in input / to protect data integrity (1) / validation is an automatic check that data is sensible / reasonable / obeys rules (1) / verification checks data has been entered/copied accurately, matching the source (1) / validation is carried out by software; verification may be manual (proofreading) or automated (double entry) (1) / neither guarantees the data is correct (1). Max 4.

8 A GP surgery receptionist enters a new patient's details from a paper registration form. Describe two verification methods that could be used. [4]

> Double entry: details are entered twice (by the same or another person) (1) / the computer compares the two and flags differences for correction (1). Proofreading: the receptionist or a colleague compares the data on screen/printout with the paper form (1) / any differences are corrected; better if done by a different person (1). Accept on-screen confirmation with the patient.

9 Explain two limitations of data validation and verification. [4]

> Any two explained, from: validation cannot detect data that is reasonable but wrong (1), e.g. 34 entered instead of 43 (1) / verification cannot detect errors in the source document (1), e.g. the customer wrote the wrong date of birth (1) / double entry can be defeated if the same error is made twice (1) and doubles workload (1) / proofreading is slow and errors are missed if rushed (1) / simple rules may accept impossible values (1), e.g. 31 April passes a 1–31 check (1)

10 A district council in Northern Ireland is replacing paper forms for bulky-waste collection requests with an online booking system. Discuss how the council can make sure that the information produced by the new system is of good quality. You should refer to the factors affecting the quality of information, validation and verification. [8]

> Indicative content: quality factors applied to the scenario – accurate (validated addresses, item descriptions), relevant (only details crews need), up to date (bookings visible immediately, cancellations removed), complete (address, item type, access notes, date all required), well presented (clear collection lists/maps for crews), reliable (data from the resident, verified). Validation: presence checks on compulsory fields; lookup check on postcode/address against council gazetteer; range check on number of items; format check on postcode and phone number; date must be a valid collection day. Verification: double entry of email; on-screen confirmation summary; confirmation email for resident to check. Limitations: residents may give wrong but valid details; validation cannot ensure correctness; staff spot checks. Levels: 7–8 well-structured discussion covering quality factors, validation and verification with specific applied examples and limitations; 4–6 covers at least two areas with some application; 1–3 lists factors or checks with little context.

E Quick quiz

Circle one answer · 9 marks

1 A school records the value 14 for a pupil. Which of these turns that data into information? [1]

A Storing 14 as an integer
B Labelling it 'Days absent this term: 14' ✓
C Typing 14 twice to check it
D Rounding 14 to 10

2 Why would validation NOT detect that a customer's age has been typed as 34 instead of 43? [1]

A Validation only runs on text fields
B Validation only checks data after it has been stored
C 34 is still a sensible, reasonable age, so it passes the rules ✓
D Ages cannot be validated

3 Typing an email address into two boxes so they can be compared is... [1]

A a format check
B a range check
C a lookup check
D verification by double entry ✓

4 Which check would ensure that a mobile number contains exactly 11 characters? [1]

A Length check ✓
B Range check
C Presence check
D Lookup check

5 A café uses last month's sales figures to order this week's stock. Which quality factor is lacking? [1]

A Presentation
B Reliability
C Up to date (currency) ✓
D Relevance

6 A club only accepts membership numbers that exist in its members file. This is a... [1]

A type check
B lookup check ✓
C format check
D proofreading check

7 Which statement about verification is true? [1]

A It checks the data matches the original source ✓
B It checks the data is within set limits
C It is always carried out automatically by software
D It guarantees the source document is correct

8 A news site reports a claim from one unverified social-media post. Which quality factor is most at risk? [1]

A Completeness
B Presentation
C Currency
D Reliability ✓

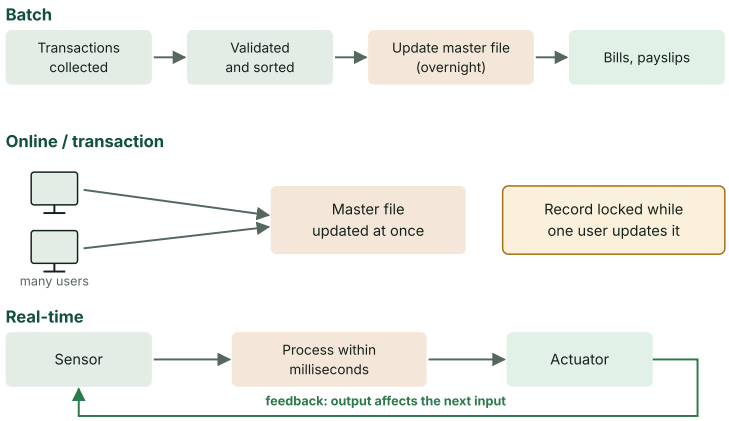
9 What does GIGO mean? [1]

A Good input gives good output
B Poor-quality input data produces poor-quality output information ✓
C Graphics in, graphics out
D Data is processed in the order it arrives

Check digits, batch totals, data capture and processing methods

FD07 ANSWERS

I can: Calculate a Modulus 11 check digit and use it to show that a transposition error has been detected · Use batch, control and hash totals to check that every document in a batch has been entered exactly once · Compare manual and automated data capture methods · Describe batch processing, real-time transaction (online) processing and real-time control, and justify which suits a scenario



Use this diagram to help you: Batch processing collects transactions and updates the master file later; transaction processing updates it immediately with record locking; real-time processing uses a feedback loop from sensors

A Key terms

Write the letter of the matching definition · 6 marks

E Check digit

A Modulus 11

F Batch total

D Control total

C Hash total

B Batch processing

A A check digit method in which each digit is multiplied by a weighting, the products are added, and the check digit is found from the remainder after dividing by 11 (0 gives 0, 1 gives X, otherwise 11 minus the remainder).

B Processing in which transactions are collected over a period in a transaction file and processed together as one job at a later time, without user intervention.

C A batch total formed by adding up a field whose total has no meaning, such as ID numbers, and comparing it with the computer total.

D A batch total formed by adding up a meaningful field on every document, such as total hours worked, and comparing it with the computer total.

E An extra digit calculated from the other digits of a code number and added to the end, so the computer can detect entry or reading errors by recalculating it.

F A validation check in which the number of documents in a batch is counted by hand and compared with the number entered by the computer.

B Fill the gaps

Use the word bank · 9 marks

weighting | multithreading | real-time | hash total | Modulus 11 | batch processing | batch total | check digit | record locking

A **check digit** is calculated from the other digits of a code number and added to the end. In the **Modulus 11** method, each digit is multiplied by a **weighting** and the remainder after dividing by 11 is used. Counting the documents in a batch gives a **batch total**, while adding up a meaningless field such as ID numbers gives a **hash total**. In **batch processing**, data is collected over time and processed together without user interaction. In online systems, **record locking** prevents two users updating the same record at the same time. Anti-lock brakes use **real-time** processing. Splitting a single program into parts that run concurrently is called **multithreading**.

C True or false?

Tick one box · 9 marks

	True	False
If the Modulus 11 remainder is 1, the check digit is X.	☑	☐
A Modulus 11 check digit proves that a number belongs to the right customer. *(It only shows the digits are consistent with each other; a valid number can still be the wrong person's.)*	☐	☑
A hash total is the sum of a field that has a meaning, such as total hours worked. *(That is a control total; a hash total is the sum of a field with no meaning, such as ID numbers.)*	☐	☑
A batch total will not detect one document missed out and another entered twice.	☑	☐
Payroll is a typical batch processing application.	☑	☐
In batch processing the master file is always up to date. *(It is only up to date immediately after a batch run.)*	☐	☑
An online cinema booking system is an example of real-time control. *(It is real-time transaction (online) processing: the master file is updated at once while the user waits, but there are no sensors or guaranteed millisecond responses as in real-time control.)*	☐	☑
Record locking is needed to prevent double-booking in transaction systems.	☑	☐
OMR is suitable for capturing handwritten essays. *(OMR only detects marks in set positions, not free text.)*	☐	☑

D Exam-style questions

36 marks

1 State the purpose of a check digit. [1]

> To detect errors when a code number is entered or scanned, e.g. transposition/transcription errors (1)

2 A library in Strabane issues borrower numbers that end in a Modulus 11 check digit. Work out the check digit for the base number 2731, showing your working. [3]

> Weightings 5, 4, 3, 2; products 10, 28, 9, 2 (1) / checksum 49; 49 ÷ 11 = 4 remainder 5 (1) / check digit 11 − 5 = 6, full number 27316 (1)

3 A library card number is stored as 1250X. Use the Modulus 11 method to show that this number has been entered incorrectly. [3]

> Weightings 5, 4, 3, 2, 1 with X = 10 (1) / products 5 + 8 + 15 + 0 + 10 = 38 (1) / 38 ÷ 11 = 3 remainder 5; remainder is not 0, so the number is invalid (1). Alternative: check digit for 1250 is 28 MOD 11 = 6, 11 − 6 = 5, not X.

4 Explain why a code number that uses a Modulus 11 check digit must be stored as a string rather than an integer. [2]

> The check digit can be the letter X (representing 10) (1) / an integer field cannot store a letter / leading zeros would also be lost (1)

5 A swimming club in Lisburn types in 30 paper entry forms for a gala. Each form contains a swimmer ID, the swimmer's name, the number of events entered and the entry fee. Explain how a batch total, a control total and a hash total could each be used to check that every form has been entered exactly once. [6]

> Batch total: count the 30 forms by hand before entry (1) / compare with the computer's count of records entered; a difference shows a form missed or entered twice (1). Control total: add up a meaningful field by hand, e.g. total entry fees or total events entered (1) / compare with the spreadsheet's sum of that field (1). Hash total: add up the swimmer IDs by hand (meaningless total) (1) / compare with the computer total; difference shows an error and, for a single missing form, identifies its ID (1)

6 Explain why a hash total can detect an error that a batch total would not. [2]

> If one document is missed and another entered twice, the count of documents stays the same so the batch total matches (1) / but the sum of unique IDs changes, so the hash total differs and the error is detected (1)

7 Compare manual keying and optical mark recognition (OMR) as methods of capturing pupils' answers to a multiple-choice test. [4]

> OMR is much faster for large numbers of sheets (1) / OMR is more accurate as there are no typing errors (1) / manual keying needs no special scanner or pre-printed forms so is cheaper for small classes (1) / OMR can only read marks in set positions and may misread faint or stray marks; keying can handle any layout (1). Max 4, must include both methods for full marks.

8 Describe how batch processing is used to produce monthly electricity bills. [4]

> Meter readings are collected over the month into a transaction file (1) / the transaction file is validated (e.g. batch/control totals) and sorted into customer order (1) / at the end of the period it is processed in one run, often overnight, with no user interaction (1) / the master file of customer accounts is updated and bills are produced/printed for every customer (1)

9 Explain how record locking prevents two customers booking the same seat in an online theatre booking system. [3]

> When a customer selects a seat, that seat's record is locked (1) / other users cannot update/book that record until the first transaction is complete or released (1) / so the seat is only sold once, preventing double-booking/lost updates (1)

10 A hospital trust in Northern Ireland uses computer systems for staff payroll, outpatient appointment booking and intensive-care patient monitoring. Evaluate the suitability of batch processing, real-time transaction (online) processing and real-time control for each of these applications. [8]

> Indicative content: Payroll – batch: paid monthly to all staff on a fixed date; large volume; same process for every employee; run overnight when the system is quiet; no need for an always-current master file; efficient. Real-time processing unnecessary and more costly. Appointment booking – real-time transaction (online) processing: patients and staff need to see available slots immediately; master file must be current to avoid double-booking; record locking needed; batch unsuitable because slots could be booked twice. Intensive-care monitoring – real-time control: sensor readings must be processed immediately with guaranteed response times; alarms must sound within seconds; life-critical, so high reliability and duplicated hardware; batch or delayed processing dangerous. Levels: 7–8 evaluates all three with justified choices linked to the features of each method and the scenario, and comments on drawbacks; 4–6 correct methods with some justification; 1–3 identifies methods with little or no justification.

E Quick quiz

Circle one answer · 9 marks

1 Using Modulus 11, the checksum for a base number is 47. What is the check digit? [1]

A 3 **B 8 ✓**

C X D 0

2 Which application best suits real-time processing? [1]

A Payroll B Monthly bank statements

C Car airbag deployment ✓ D Exam result processing

3 The total of all the staff ID numbers on a batch of timesheets is a... [1]

A batch total B control total

C check digit **D hash total ✓**

4 What does record locking prevent? [1]

A Viruses **B Two users updating the same record at once ✓**

C Data being deleted by backups D Unauthorised logins

5 What are the Modulus 11 weightings for the four-digit base number 9150? [1]

A 5, 4, 3, 2 ✓ B 1, 2, 3, 4

C 4, 3, 2, 1 D 2, 3, 4, 5

6 Which is a feature of batch processing? [1]

A The user waits for each result **B Processing happens without user interaction ✓**

C Response in milliseconds D It relies on record locking

7 A web browser downloads images while the user scrolls. This is an example of... [1]

A multithreading ✓ B batch processing

C OMR D multi-user

8 Which data capture method suits a lottery ticket? [1]

A Keying B Voice input

C OMR ✓ D OCR of handwriting

9 Which mistake would a batch total NOT detect? [1]

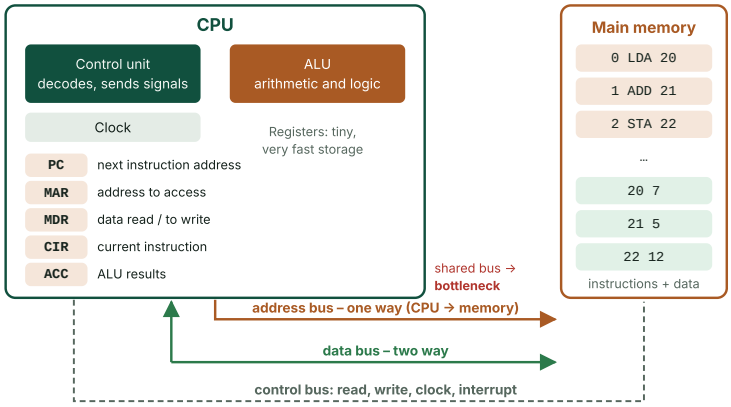
A One form missed out B One form entered twice

C One form missed and another entered twice ✓ D Two forms missed out

Internal components: CPU, registers, buses, I/O controllers and the Von Neumann architecture

FD08 ANSWERS

I can: Describe what the control unit, ALU, clock and registers each do · Explain the job of each special-purpose register and each bus, including the direction it carries signals · Calculate maximum addressable memory, or a missing bus width, showing the working in powers of two · Explain why peripherals connect through I/O controllers and ports rather than directly to the system bus · Describe the stored-program concept and the Von Neumann bottleneck



Use this diagram to help you: The Von Neumann architecture: a CPU containing the control unit, ALU, clock and registers, connected to a single main memory holding instructions and data by the address, data and control buses

A Key terms

Write the letter of the matching definition · 6 marks

E Arithmetic logic unit (ALU)

D Control unit (CU)

F Register

A Program counter (PC)

C Memory address register (MAR)

B Memory data register (MDR)

A A register that holds the address of the next instruction to be fetched; it is incremented after each fetch.

B A register that holds data or an instruction just read from memory, or data waiting to be written to memory.

C A register that holds the address of the memory location currently being read from or written to.

D The part of the CPU that fetches and decodes instructions and sends control signals to coordinate the processor, memory and I/O.

E The part of the CPU that performs arithmetic operations, logical operations and comparisons on data held in registers.

F A single, very fast storage location inside the CPU that holds data, an instruction or an address during processing.

B Fill the gaps

Use the word bank · 9 marks

control bus | arithmetic logic unit | I/O controller | memory address register | control unit | program counter | Von Neumann | address bus | data bus

In the **Von Neumann** architecture, instructions and data are stored together in main memory. The **control unit** fetches and decodes instructions, while the **arithmetic logic unit** performs calculations and comparisons. The **program counter** holds the address of the next instruction, and the **memory address register** holds the address about to be accessed. Addresses travel on the **address bus**, which is unidirectional, while data travels on the bidirectional **data bus**. Signals such as memory read and interrupt requests use the **control bus**. Peripherals are connected through an **I/O controller**, which converts voltages, protocols and data formats.

C True or false?

Tick one box · 9 marks

	True	False
The data bus is bidirectional.	☑	☐
The MAR holds the instruction currently being executed. *(The CIR holds the current instruction; the MAR always holds a memory address.)*	☐	☑
In the Von Neumann architecture, data and instructions are stored in the same memory.	☑	☐
The control unit performs arithmetic calculations. *(The ALU performs calculations; the control unit fetches, decodes and coordinates.)*	☐	☑
Adding one line to the address bus doubles the number of addressable locations.	☑	☐
The control bus carries memory addresses. *(Addresses use the address bus; the control bus carries signals such as read, write, clock and interrupt.)*	☐	☑
All data must be in a register before the ALU can process it.	☑	☐
A 10-bit address bus can address 2048 locations. *(2^{10} = 1024 locations (0 to 1023); 2048 locations would need 11 address lines.)*	☐	☑
An I/O controller can buffer data so that a fast processor is not held up by a slow peripheral.	☑	☐

D Exam-style questions

30 marks

1 State two tasks carried out by the control unit. [2]

> Any two from: fetches instructions from memory, decodes and executes them (fetch–execute cycle), using the ALU as needed (1) / sends control signals to coordinate and monitor hardware so commands are carried out (1) / controls the input and output of data / movement of data into and out of the processor (1)

2 Describe the purpose of the memory data register (MDR). [2]

> Holds data/an instruction that has just been read (fetched) from memory (1) / or data that is waiting to be written to memory (1) – acts as a buffer between the CPU and memory

3 Explain why the address bus is unidirectional but the data bus is bidirectional. [2]

> Only the CPU sends addresses / memory and I/O never need to send an address back, so the address bus only carries signals from CPU to memory/I/O (1) / data must travel both to and from the CPU (reads and writes), so the data bus is two-way (1)

4 Describe how the processor writes a value to main memory, referring to the address, data and control buses. [4]

The address of the location is placed on the address bus (from the MAR) (1) / the data to be stored is placed on the data bus (from the MDR) (1) / a write signal is sent on the control bus (1) / memory stores the data from the data bus in the addressed location (overwriting its contents) (1)

5 A self-service checkout in a Newry supermarket has a 22-bit address bus and a 32-bit data bus. Calculate the maximum addressable memory in MB. Show your working. [3]

2^22 × 32 bits = 2^22 × 2^5 = 2^27 bits (1) / ÷ 2^3 = 2^24 bytes (1) / ÷ 2^20 = 2^4 = 16 MB (1)

6 A gaming PC in an Armagh youth club has 4 GB of RAM and a 32-bit data bus. Calculate the width of the address bus needed. Show your working. [3]

4 GB in bits = 2^2 × 2^30 × 2^3 = 2^35 bits (1) / 2^(width) × 2^5 = 2^35, so 2^(width) = 2^30 (1) / address bus width = 30 bits (1)

7 The controller in a home heating system has 12 address lines but its program needs 6000 memory locations. Explain whether the address bus is wide enough. [2]

12 lines address 2^12 = 4096 locations (1) / 4096 < 6000 so it is not wide enough; locations above 4095 could not be addressed – at least 13 lines (8192) needed (1)

8 Explain two reasons why a printer is connected to a computer through an I/O controller rather than directly to the system bus. [4]

Any two explained, e.g.: devices operate at different speeds (1), so the controller buffers data so the CPU is not held up (1) / devices use different voltages (1), so the controller converts voltage levels to suit the computer (1) / devices use different protocols/command formats (1), so the controller performs protocol conversion (1) / data formats differ (1), e.g. converts a 16-bit value to the 32-bit form required (1) / impractical to design a CPU able to control every peripheral directly (1)

9 Explain why most computers are not fitted with the maximum amount of memory their address bus could address. [2]

Any two from: RAM is expensive (1) / most programs/users do not need the maximum amount (1) / some addresses are allocated to I/O ports so cannot be used for RAM (1)

10 A technology club at a school in Ballymena is building a model to show younger pupils how a PC works. Discuss how the Von Neumann architecture organises the components of a computer, including the roles of the registers and the system bus, and the main limitation of this design. [6]

Indicative content: instructions and data stored together in the same main memory in binary (stored-program concept); single processor fetches and executes instructions sequentially; control unit fetches/decodes/coordinates; ALU performs arithmetic and logic; clock synchronises; PC holds next instruction address, MAR address being accessed, MDR data read/written, CIR current instruction, ACC results; address bus one-way CPU to memory, data bus two-way, control bus carries read/write/interrupt signals; I/O devices via controllers; secondary storage for permanent storage; limitation: instructions and data share one bus so cannot be fetched simultaneously – Von Neumann bottleneck, CPU waits for memory. Levels: 5–6 marks: thorough, accurate description of components, registers and all three buses linked to how they work together, plus clear explanation of the bottleneck, good technical vocabulary. 3–4 marks: several components and buses correctly described, limitation stated but not fully explained. 1–2 marks: limited, list-like points with some inaccuracies.

E Quick quiz

Circle one answer · 9 marks

1 Which register holds the address of the next instruction to be fetched? [1]

A Memory address register (MAR)

B Current instruction register (CIR)

C Program counter (PC) ✓

D Accumulator (ACC)

2 A games console's processor has 32 address lines and a 64-bit data bus. What is the most memory it can address? [1]

A 4 GB

B 32 GB ✓

C 64 GB
D 256 GB

3 Which bus carries signals such as 'memory write' and 'interrupt request'? [1]

A Control bus ✓
B Data bus
C Address bus
D Expansion bus

4 How many locations can a 16-bit address bus access? [1]

A 16
B 256
C 32 768
D 65 536 ✓

5 What is the Von Neumann bottleneck? [1]

A Running out of RAM
B Instructions and data competing for one shared bus ✓
C The processor overheating
D Having too few registers

6 Which register acts as a two-way buffer between the processor and memory? [1]

A CIR
B PC
C MDR ✓
D MAR

7 Which is NOT a task of an I/O controller? [1]

A Voltage conversion
B Decoding program instructions ✓
C Buffering data
D Protocol conversion

8 A computer has 16 GB of RAM and a 64-bit data bus. What is the address bus width? [1]

A 31 bits ✓
B 32 bits
C 34 bits
D 28 bits

9 Which component decodes instructions? [1]

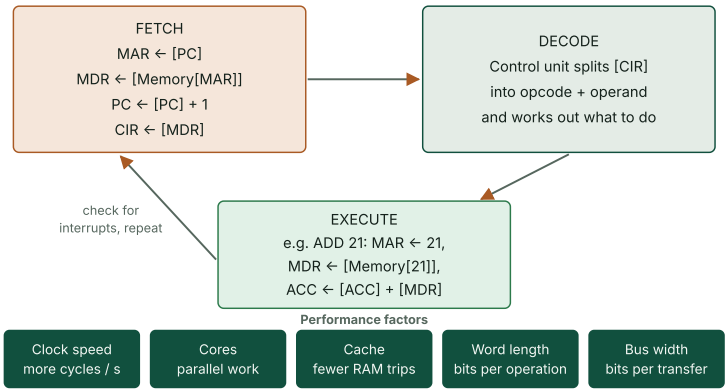
A ALU
B Clock
C MDR
D Control unit ✓

AS 2: FUNDAMENTALS OF DIGITAL TECHNOLOGY · ARCHITECTURE

The fetch–decode–execute cycle, pipelining and processor performance

FD09 ANSWERS

I can: Describe each stage of the fetch–decode–execute cycle in the correct order, naming the registers and buses used · Draw a pipelining diagram and work out how many time units a set of instructions needs · Explain how clock speed, cores, word size, cache, bus width and RAM affect processing speed, and the limits of each · Interpret a processor specification and justify why clock speed alone does not decide performance



Use this diagram to help you: The fetch-decode-execute cycle shown as a loop with the register transfers at each stage, and the factors that affect processor performance

A Key terms

Write the letter of the matching definition · 6 marks

E Fetch–decode–execute cycle

A Opcode

F Operand

D Instruction set

B Pipelining

C Clock speed

A The part of a machine-code instruction that specifies the operation to be performed.

B Overlapping the fetch, decode and execute stages of successive instructions so that several instructions are being processed at once.

C The number of clock cycles the processor carries out per second, measured in hertz (e.g. 3.5 GHz).

D The complete set of machine-code instructions that a particular processor is designed to recognise and execute.

E The continuous process by which the CPU fetches an instruction from memory, decodes it and carries it out.

F The part of a machine-code instruction that holds the data to be used or the address where the data is stored.

B Fill the gaps

Use the word bank · 9 marks

clock speed | opcode | control unit | program counter | benchmark | incremented | accumulator | memory data register | pipelining

In the fetch stage the address in the **program counter** is copied into the MAR, and the PC is **incremented**. The instruction arrives in the **memory data register** along the data bus and is copied into the CIR. The **control unit** then decodes the instruction into an **opcode** and an operand. During

the execute stage, calculations are carried out by the ALU and the result is stored in the **accumulator**. Overlapping the stages of different instructions is called **pipelining**. A processor's **clock speed** is measured in hertz, but a fairer comparison uses a **benchmark** reported in MIPS or FLOPS.

C True or false?

Tick one box · 9 marks

	True	False
During the fetch stage, the PC is incremented.	☑	☐
The ALU decodes each instruction. *(The control unit decodes instructions; the ALU performs calculations and comparisons.)*	☐	☑
Doubling the number of cores always doubles performance. *(Only work that can be split into parallel threads benefits, and cores share cache and memory.)*	☐	☑
With pipelining, three instructions using fetch, decode and execute can be completed in five time units.	☑	☐
Higher clock speeds produce more heat.	☑	☐
The CIR holds the address of the next instruction. *(The PC holds that address; the CIR holds the instruction currently being decoded and executed.)*	☐	☑
64-bit software will run on a 32-bit processor. *(32-bit software runs on a 64-bit processor, but not the other way round.)*	☐	☑
FLOPS measures the number of floating-point operations a processor carries out per second.	☑	☐
A 2 GHz processor carries out 2 billion clock cycles per second.	☑	☐

D Exam-style questions

35 marks

1 State the three stages of the fetch–execute cycle. [1]

> Fetch, decode, execute (all three for 1 mark)

2 Describe the fetch stage of the fetch–decode–execute cycle, naming the registers and buses involved. [6]

> The address in the program counter (PC) is copied into the memory address register (MAR) (1) / the PC is incremented to point to the next instruction (1) / the address held in the MAR is put onto the address bus (1) / a read signal is sent on the control bus (1) / the instruction at that address is sent along the data bus into the memory data register (MDR) (1) / the instruction is copied from the MDR into the current instruction register (CIR) (1)

3 Explain what happens during the decode stage, referring to the opcode and operand. [3]

> The control unit interprets the instruction in the CIR (1) / it splits it into the opcode, which identifies the operation to perform (1) / and the operand, which is the data or the address of the data (1) / (accept: prepares registers for execution)

4 Explain how a jump instruction changes the normal sequence of the fetch–execute cycle. [2]

> The jump loads a new address into the program counter (1) / so the next instruction fetched comes from that address rather than the next sequential location (1)

5 Using a diagram, show how four instructions can be completed in the minimum number of time units using pipelining, and state how many time units are saved compared with no pipelining. [4]

> Diagram with time units T1–T6 and four instructions staggered by one unit, each F, D, E (2: 1 for correct staggering, 1 for correct stage order) / pipelined total 6 time units (1) / non-pipelined 12, so 6 time units saved (1)

6 A processor in a ticket machine at Lanyon Place station runs at 1.5 GHz. Calculate how many clock cycles it performs in one second and the time taken for one cycle in nanoseconds. [2]

> 1.5 × 10^9 = 1 500 000 000 cycles per second (1) / one cycle = 1 ÷ 1.5 × 10^9 s ≈ 0.67 ns (1)

7 Explain why clock speed alone is not a good measure of processor performance. [2]

> Processors with different architectures do different amounts of work per clock cycle (1) / other factors such as number of cores, cache size, pipelining and word size also affect speed (1)

8 An architect's office in Enniskillen is replacing a 32-bit computer with a 64-bit one. Explain two ways this could improve performance. [4]

> Any two explained, e.g.: a 64-bit processor handles 64 bits of data in one operation (1) so large values/calculations need fewer operations (1) / 64-bit addressing allows more than 4 GB of RAM to be used (1) so large CAD files can stay in RAM, reducing slow swapping to disk (1) / wider registers and data paths move more data per operation (1)

9 Explain why a quad-core processor may not run a program four times faster than a single-core processor. [3]

> Not all of the program can be split into parallel tasks; some parts must run sequentially (1) / the software must be written to use multiple threads (1) / cores share cache, memory and the bus, so they may wait for each other (1) / there is overhead in dividing work and combining results (1) (max 3)

10 A sports club in Portadown wants a new computer for editing match videos and running its membership database. Evaluate how clock speed, number of cores, cache size and amount of RAM would each influence the speed of processing, and recommend which matters most for this club. [8]

> Indicative content: clock speed – more cycles per second so more instructions executed, benefits single-threaded tasks such as some database queries, but more heat/power and only fair to compare similar architectures; cores – video editing/rendering is multi-threaded so many cores give large gains, database may not use many cores, limited by sequential parts; cache – keeps frequently used instructions/data near CPU, reduces RAM access, benefit for repeated operations, expensive; RAM – video files are large, enough RAM avoids virtual memory swapping which would slow processing badly; database is small. Recommendation: e.g. cores and RAM matter most for video editing, justified by the workload; reasoned conclusion. Levels: 7–8 marks: all four factors explained accurately with limitations, applied to both uses, justified recommendation. 4–6 marks: most factors explained with some application to the club, recommendation given with partial justification. 1–3 marks: factors described generically or listed, little evaluation.

E Quick quiz

Circle one answer · 9 marks

1 In the fetch stage, what happens immediately after the contents of the program counter are copied into the MAR? [1]

A The instruction is decoded

B The PC is incremented to point to the next instruction ✓

C The ALU adds the operand to the accumulator

D The CIR is cleared

2 Why can a 2.8 GHz processor sometimes outperform a 3.4 GHz processor? [1]

A Lower clock speeds always run cooler and so faster

B It may complete more instructions per cycle, or have more cores and cache ✓

C Clock speed is measured differently on each chip

D The 3.4 GHz chip must be 32-bit

3 Which register receives the instruction from memory via the data bus? [1]

A MAR

B PC

C MDR ✓

D ACC

4 A 3.5 GHz processor performs how many cycles per second? [1]

A 3.5 million
B 350 million
C 3.5 billion ✓
D 35 billion

5 How many time units do five instructions need with a three-stage pipeline? [1]

A 7 ✓
B 5
C 15
D 10

6 During decoding, what is the opcode? [1]

A The address of the next instruction
B The data value used by the instruction
C The register that stores the result
D The part of the instruction that says which operation to perform ✓

7 Which change would most help a computer that keeps swapping data to virtual memory? [1]

A A faster clock
B More cores
C More RAM ✓
D A wider address bus

8 What does a jump instruction change during execution? [1]

A The program counter ✓
B The accumulator
C The MDR
D The clock speed

9 Which benchmark measure is most relevant for a scientific simulation using real numbers? [1]

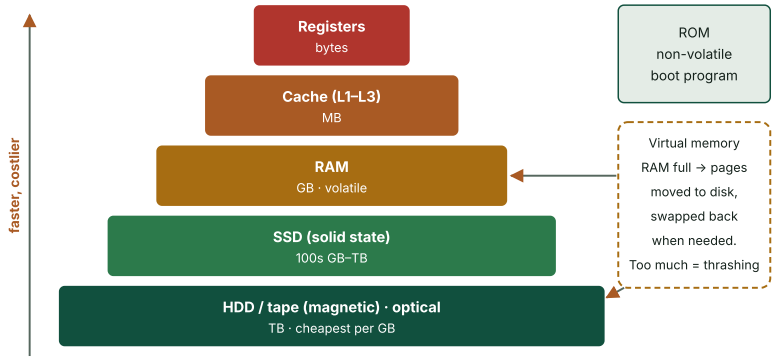
A MIPS
B Hz
C Word size
D FLOPS ✓

AS 2: FUNDAMENTALS OF DIGITAL TECHNOLOGY · ARCHITECTURE

Main memory, cache and secondary storage

FD10 ANSWERS

I can: Compare RAM, ROM and cache in terms of volatility, speed, size and use · Explain how the cache controller uses cache to reduce waiting time, and how virtual memory works · Explain why secondary storage is needed and describe how magnetic, optical and flash media store data · Recommend and justify a storage medium for a given scenario using speed, capacity, durability, portability, reliability and cost



Use this diagram to help you: The memory hierarchy from registers to magnetic and optical storage: faster and more expensive at the top, larger at the bottom, with virtual memory linking RAM and secondary storage

A Key terms

Write the letter of the matching definition · 6 marks

E Main memory

C RAM

F ROM

A Volatile

D Cache memory

B Cache controller

A Describes memory that loses its contents when the power is switched off.

B The device that transfers data from RAM into cache, predicting and pre-fetching the data the processor is likely to need next.

C Random access memory: volatile read/write main memory that holds the operating system, programs and data currently in use.

D Small, very fast and expensive memory on or beside the processor that holds frequently used data and instructions and data read ahead from RAM.

E The memory that the processor can address directly, holding the programs and data currently in use (RAM and ROM).

F Read-only memory: non-volatile main memory whose contents are fixed when made, used for start-up firmware and embedded programs.

B Fill the gaps

Use the word bank · 9 marks

cache controller | RAM | thrashing | virtual memory | serial | ROM | SSD | cache | tracks

RAM is volatile, so its contents are lost when the power is switched off, whereas **ROM** keeps the start-up firmware without power. Frequently used instructions are kept in **cache** memory, and the **cache controller** reads ahead to pre-fetch data. When RAM is full the operating

system uses **virtual memory** on secondary storage, and excessive swapping is called disk **thrashing**. A hard disk divides each platter into **tracks** and sectors, while magnetic tape gives only **serial** access. An **SSD** has no moving parts, giving fast access and low power use.

C True or false?

Tick one box · 9 marks

	True	False
Cache memory is non-volatile. *(Cache is volatile; its contents are lost when the power is removed.)*	☐	☑
Virtual memory allows more programs to be open than RAM alone would allow.	☑	☐
Optical discs generally have a higher capacity than hard disk drives. *(Hard disks hold terabytes; even a four-layer Blu-ray disc holds only 128 GB.)*	☐	☑
SSDs are more resistant to physical shock than hard disk drives.	☑	☐
Magnetic tape provides direct access to any file. *(Tape is serial access: the drive must wind past earlier data to reach a file.)*	☐	☑
Adding more RAM usually increases the use of virtual memory. *(More RAM means the computer needs to swap pages to disk less often.)*	☐	☑
A CD-R can be written to once but read many times.	☑	☐
Level 1 cache is usually smaller and faster than level 3 cache.	☑	☐
Hard disks usually have a lower cost per gigabyte than SSDs.	☑	☐

D Exam-style questions

37 marks

1 State two differences between RAM and ROM. [2]

> Any two from: RAM is volatile, ROM is non-volatile (1) / RAM can be read and written, ROM can only be read (in normal use) (1) / RAM holds programs and data currently in use, ROM holds fixed start-up programs/firmware (1) / RAM is usually much larger than ROM (1)

2 Explain why the start-up instructions of a PC are stored in ROM rather than RAM. [2]

> RAM is volatile so it is empty when the computer is switched on (1) / ROM is non-volatile so the start-up instructions are available immediately to load the operating system (1)

3 Using a diagram, explain how the CPU makes use of cache memory when it needs an instruction. [4]

> Diagram showing CPU, cache, cache controller and RAM (1) / CPU first checks the cache for the instruction (1) / if present (cache hit), it is used directly, having been pre-fetched by the cache controller (1) / if not (miss), it is fetched from RAM, placed in cache and then processed; controller may read further data ahead (1)

4 Describe how virtual memory is used when RAM is full, and explain one effect on performance. [4]

> RAM is full, so the OS uses part of secondary storage as virtual memory (1) / pages not currently needed are moved from RAM to the disk (1) / when needed they are swapped back into RAM (1) / performance falls because secondary storage is much slower than RAM / excessive swapping causes disk thrashing (1)

5 State three benefits of secondary storage. [3]

> Capacity – much larger than RAM (1) / portability – data can be moved between locations (1) / security – backup copies can be kept (1) / (accept: non-volatile, keeps data when power is off)

6 Describe the steps involved in writing a file to a hard disk. [4]

> Platters spin around the spindle (1) / data is sent to the hard disk via the disk controller (1) / the arm positions the read/write head over the required track (1) / when the required sector passes under the head the data is written by magnetising the surface (1) (accept: OS checks file table for free space)

7 A computer in a Coleraine estate agent's office has a CD-ROM drive. Explain why staff cannot use it to back up their files. [2]

> A CD-ROM drive is read only (1) / it cannot write ('burn') data to a disc, so a CD-R/RW writer would be needed (1)

8 A wildlife film-maker records footage on cameras in the Mourne Mountains. Explain two reasons, other than cost, why the cameras use flash memory cards. [4]

> Any two explained, e.g.: no moving parts (1) so robust/not damaged by knocks or vibration outdoors (1) / small and lightweight (1) so suits a portable camera (1) / low power use (1) so camera batteries last longer (1) / fast write speed (1) so HD video can be recorded without loss (1) / removable (1) so cards can be swapped when full (1)

9 For each of the following, recommend a different storage medium and justify your choice: (a) a hospital trust archiving 40 TB of scans that are rarely viewed; (b) a teacher carrying coursework between school and home; (c) a band selling copies of its album at gigs. [6]

> (a) Magnetic tape (1) – huge capacity at very low cost per GB; serial access does not matter as scans are rarely viewed (1) / (b) USB flash drive (1) – small, portable, rewritable and robust for carrying daily (1) / (c) optical disc e.g. CD-ROM/CD-R (1) – cheap to produce, light, flat to carry/sell, contents cannot be altered (1). Different device and reason each time.

10 A district council is replacing the hard disk drives in its staff laptops with solid-state drives. Evaluate this decision. [6]

> Indicative content – advantages: faster access/boot and application loading (no moving parts, no seek time), so staff more productive; lower power use so longer battery life; more robust to knocks when laptops are carried to meetings/site visits; lighter, smaller, silent; less heat. Disadvantages: higher cost per GB, so either more expensive or smaller capacity; limited write cycles (mitigated by wear levelling); data recovery after failure can be harder; cost of migration. Conclusion: for mobile staff the benefits outweigh cost, especially as prices fall. Levels: 5–6 marks: balanced evaluation with several advantages and disadvantages explained in the council context and a justified conclusion. 3–4 marks: advantages and disadvantages described with limited context or conclusion. 1–2 marks: one-sided or list of points.

E Quick quiz

Circle one answer · 9 marks

1 Why must the instructions that start a PC be held in ROM rather than RAM? [1]

A ROM is faster than RAM

B RAM is empty when the computer is switched on, but ROM keeps its contents ✓

C ROM has a larger capacity than RAM

D RAM cannot hold program instructions

2 A delivery company's drivers record signatures on handheld devices that are often dropped. Which storage is most suitable? [1]

A Hard disk drive

B Magnetic tape

C Flash memory ✓

D DVD-RW

3 What is disk thrashing? [1]

A Defragmenting a disk

B Excessive swapping of pages between RAM and disk ✓

C A crashed hard disk head

D Formatting a disk

4 Which storage is best for distributing a film to thousands of customers cheaply? [1]

A SSD
B Magnetic tape
C Hard disk
D Optical disc ✓

5 Which is fastest? [1]

A RAM
B L1 cache ✓
C SSD
D Hard disk

6 What does it mean when the cache controller 'reads ahead'? [1]

A It deletes data the CPU has finished with
B It pre-fetches the data it predicts will be needed next ✓
C It copies the cache to secondary storage
D It reads the next file from the hard disk

7 Which is a disadvantage of SSDs compared with hard disks? [1]

A Higher cost per gigabyte ✓
B Moving parts
C Slower access
D Higher power use

8 Which medium is most suitable for archiving 50 TB of records that are rarely accessed? [1]

A USB flash drive
B Blu-ray disc
C SSD
D Magnetic tape ✓

9 On a hard disk, what are tracks? [1]

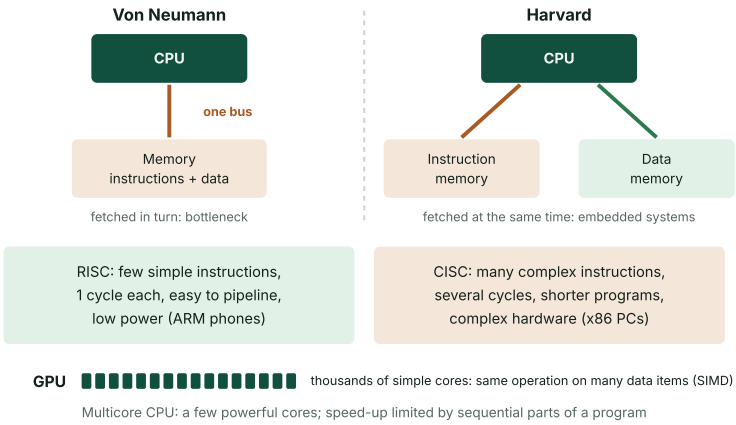
A Concentric circles on each platter surface ✓
B Pie-shaped wedges of a disc
C The arms that hold the heads
D The cables to the motherboard

AS 2: FUNDAMENTALS OF DIGITAL TECHNOLOGY · ARCHITECTURE

Alternative architectures: parallel, multicore, Harvard, RISC and CISC, GPUs

FD11 ANSWERS

I can: Explain how parallel processing and multi-core processors increase throughput and why the gains are limited · Compare Harvard and Von Neumann architectures, giving typical uses of each · Compare RISC and CISC processors and link their features to power use, pipelining and program size · Explain why GPUs suit graphics, machine learning and other highly parallel tasks, and choose an architecture for a scenario



Use this diagram to help you: Von Neumann architecture with one shared memory and bus compared with Harvard architecture with separate instruction and data memories, plus RISC and CISC and GPU summaries

A Key terms

Write the letter of the matching definition · 6 marks

E Parallel processing

C Multi-core processor

F Thread

D Harvard architecture

A Embedded system

B RISC

A A computer system built into a larger device to perform one dedicated function, such as a heating controller.

B Reduced instruction set computer: a processor with a small set of simple, fixed-length instructions, most executed in one clock cycle.

C A single chip containing two or more complete processing cores that can each execute instructions at the same time.

D A computer design with separate memories and buses for instructions and data, so both can be accessed at the same time.

E The simultaneous use of two or more processors or cores to work on different parts of the same task.

F A sequence of instructions within a program that can be scheduled to run independently, for example on a separate core.

B Fill the gaps

Use the word bank · 9 marks

embedded systems | parallel processing | SIMD | multi-core | sequential | RISC | Harvard | GPU | CISC

Using several processors or cores to work on parts of a task at the same time is called **parallel processing**. A **multi-core** processor has several complete cores on one chip, but any **sequential** part of a program limits the speed-up. The **Harvard** architecture has separate memories and buses for instructions and data, and is common in **embedded systems**. A **RISC** processor has a small set of simple instructions, most taking one clock cycle, while most desktop PCs use x86 processors, which are **CISC**. A **GPU** contains thousands of simple cores that apply the same instruction to many data items, an approach called **SIMD**.

C True or false?

Tick one box · 9 marks

	True	False
Harvard architecture allows an instruction and data to be fetched at the same time.	☑	☐
RISC processors usually use more power than CISC processors. *(RISC needs simpler circuitry with fewer transistors, so it generally uses less power.)*	☐	☑
Programs compiled for RISC often contain more instructions than the same program compiled for CISC.	☑	☐
A GPU is better than a CPU for every type of task. *(GPUs are poor at sequential, branching tasks such as running the operating system.)*	☐	☑
ARM processors in smartphones are an example of RISC.	☑	☐
Adding more cores always makes a program run faster. *(Only parallelisable parts benefit; sequential sections and overheads limit the gain.)*	☐	☑
Most desktop PCs use a pure Harvard architecture. *(PCs use Von Neumann main memory, with Harvard-style split L1 caches inside each core.)*	☐	☑
SIMD means a single instruction is applied to multiple data items at the same time.	☑	☐
CISC instruction sets often include instructions of variable length.	☑	☐

D Exam-style questions

33 marks

1 Describe what is meant by parallel processing. [2]

> Two or more processors/cores are used at the same time (1) / to work on different parts of the same task/program, which is split into parts (1)

2 Explain two limitations of adding more cores to a processor. [4]

> Any two explained, e.g.: some parts of a program are sequential (1) so they cannot be split and still run on one core, limiting speed-up (1) / software must be written to use multiple threads (1) otherwise extra cores are unused (1) / cores share cache, memory and bus (1) so they may wait for each other (1) / overhead of splitting work and combining results (1) reduces the gain (1) / more cores increase cost and power use (1)

3 A data-analysis job takes 60 seconds on one core. 45 seconds of the work can be run in parallel and the rest cannot. Calculate how long the job takes on 3 cores, and state the shortest time it could ever take however many cores are added. [3]

> Sequential part = 60 − 45 = 15 s (1) / on 3 cores: 15 + 45 ÷ 3 = 15 + 15 = 30 s (1) / shortest possible time = 15 s (the sequential part) (1)

4 Describe two differences between the Harvard and Von Neumann architectures. [4]

> Any two pairs: Von Neumann has one shared memory for instructions and data, Harvard has separate memories (2) / Von Neumann uses one set of buses, Harvard separate buses for instructions and data (2) / Von Neumann fetches instructions and data one after another, Harvard can fetch both at once (2) / Harvard memories can differ in size/word length/type, Von Neumann memory is uniform and flexible (2)

5 State one type of device that commonly uses the Harvard architecture. [1]

> Any one from: embedded system / microcontroller / digital signal processor / named embedded device e.g. washing machine controller (1)

6 A company in Craigavon makes smart thermostats. Explain why the thermostat's microcontroller is likely to use a Harvard architecture. [3]

> The thermostat runs one fixed program that can be held in separate non-volatile instruction memory (1) / sensor readings and settings held in a small separate data RAM (1) / instructions and data fetched at the same time over separate buses, giving fast and predictable response (1) / microcontrollers are cheap and low power (1) (max 3)

7 Explain why RISC processors are commonly used in smartphones. [3]

> Simpler instruction set needs fewer transistors/simpler circuitry (1) / so uses less power, giving longer battery life (1) / and produces less heat, so no fan is needed in a thin phone (1) / single-cycle fixed-length instructions pipeline efficiently, giving good performance (1) (max 3)

8 Explain why GPUs are used to train machine-learning models. [3]

> GPUs have hundreds/thousands of simple cores (1) / training involves huge numbers of identical calculations (e.g. matrix multiplications) on large data sets (1) / the GPU applies the same instruction to many data items in parallel (SIMD), so training is much faster than on a CPU (1)

9 State what is meant by a co-processor and give one example. [2]

> An additional processor that takes over a specific type of task from the main CPU (1) / e.g. GPU, sound processor, maths co-processor, neural engine (1)

10 Compare RISC and CISC processors, referring to their instruction sets, use of clock cycles, hardware complexity and typical uses. [8]

> Indicative content: instruction set – RISC small set of simple, fixed-length instructions; CISC large set of complex, variable-length instructions, some doing several operations; clock cycles – RISC most instructions one cycle; CISC complex instructions several cycles; program length – RISC more instructions, compiler does more work; CISC shorter programs, less memory for code; hardware – RISC simpler, fewer transistors, more registers, lower power and heat, easy to pipeline; CISC more complex circuitry, harder to pipeline, more power; typical uses – RISC: ARM in phones, tablets, embedded devices, some laptops; CISC: x86 in desktop PCs and servers, backward compatibility. Credit modern convergence (x86 decodes to micro-operations). Levels: 7–8 marks: detailed paired comparison across all four areas with accurate consequences and examples. 4–6 marks: comparison of most areas, some consequences, may be partly one-sided. 1–3 marks: isolated facts about one or both, little comparison.

E Quick quiz

`Circle one answer · 9 marks`

1 Why does a smartphone usually use a RISC processor rather than a CISC one? [1]

A RISC chips have more complex instructions

B RISC chips use simpler circuitry, so they need less power and produce less heat ✓

C RISC chips cannot run apps

D RISC chips use the Von Neumann bottleneck

2 A task takes 100 s on one core, and 40 s of it must run sequentially. What is the shortest time it could take with unlimited cores? [1]

A 0 s

B 10 s

C 40 s ✓

D 60 s

3 Which architecture uses separate memories for instructions and data? [1]

A Von Neumann
B Harvard ✓
C CISC
D Multi-core

4 Which is a characteristic of RISC? [1]

A Variable-length complex instructions
B Many cycles per instruction
C Few registers
D Simple instructions that are easy to pipeline ✓

5 Why are GPUs suited to image rendering? [1]

A They have more cache per core than CPUs
B They use the Harvard architecture
C The same calculation is applied to many pixels in parallel ✓
D They are CISC processors

6 Which device most likely uses a Harvard microcontroller? [1]

A Microwave oven controller ✓
B Web server
C Desktop PC
D Cloud database

7 Most desktop PCs use which instruction set approach? [1]

A RISC only
B CISC (x86) ✓
C Harvard
D SIMD

8 What is a co-processor? [1]

A A backup CPU used if the main one fails
B A second operating system
C A processor that takes over a specific type of work from the CPU ✓
D A register inside the ALU

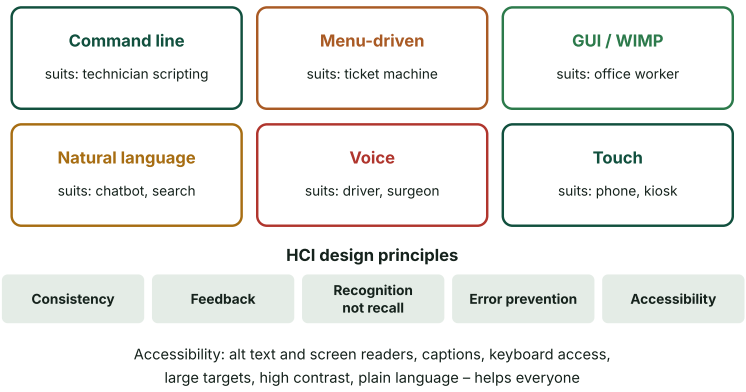
9 Which statement about multi-core processors is correct? [1]

A Each core can execute its own stream of instructions at the same time ✓
B All cores must run the same instruction
C Single-threaded programs use every core
D They remove the need for cache

User interfaces and HCI design

FD12 ANSWERS

I can: Describe the qualities of a good user interface and apply HCI principles such as consistency and feedback · Compare WIMP, command line and forms dialogue interfaces and justify a choice for given users and contexts · Explain how text, sound and video are input on PCs, laptops, tablets and smartphones · Explain how interfaces can be made accessible to users with a range of needs



Use this diagram to help you: Six types of user interface with an example user for each, and the key HCI design principles

A Key terms

Write the letter of the matching definition · 6 marks

B User interface

E Graphical user interface (GUI)

F Command line interface (CLI)

C Forms dialogue interface

D Menu-driven interface

A Voice (dialogue) interface

A A user interface that is controlled by spoken commands, using speech recognition.

B The means by which a user and a computer communicate, including what is displayed and how the user gives input.

C A user interface that guides the user through a structured on-screen form using text boxes, drop-down lists, check boxes and radio buttons.

D A user interface that presents the user with a limited list of options to choose from, often in a hierarchy.

E A user interface that uses windows, icons, menus and a pointer (WIMP) to allow the user to interact visually.

F A user interface in which the user types commands in an exact syntax at a prompt for the computer to carry out.

B Fill the gaps

Use the word bank · 9 marks

user interface | feedback | forms dialogue | accessibility | WIMP | command line | pointer | switches | consistency

The part of a system through which a person and a computer exchange instructions and results is the **user interface**. A graphical interface is built from windows, icons, menus and a **pointer**, which is why it is called **WIMP**. A **command line** interface requires users to type exact commands, which may include **switches** to change their behaviour. A **forms dialogue** interface uses text boxes, drop-

down lists and radio buttons, and can validate data as it is entered. Using the same layout and colours throughout is called **consistency**, and showing a progress bar is an example of **feedback**. Providing alt text and captions improves **accessibility**.

C True or false?

Tick one box · 9 marks

	True	False
Command line interfaces need fewer system resources than GUIs.	☑	☐
A forms dialogue interface is ideal for experts who want to work as quickly as possible. *(Forms suit beginners; experts can find them slow and tedious because they are limited to the options provided.)*	☐	☑
The P in WIMP stands for program. *(It stands for pointer – the symbol moved by a mouse or finger to select items.)*	☐	☑
Voice interfaces can help users with limited mobility.	☑	☐
Using colour alone to show errors is good accessible design. *(Colour-blind users may not see the difference; add text or icons as well.)*	☐	☑
Data can be validated as it is entered in a forms dialogue interface.	☑	☐
A forgiving interface lets users undo mistakes easily.	☑	☐
Biometric interfaces raise no privacy concerns because the data is part of the body. *(Stored biometric data is highly sensitive personal data and cannot be changed if it is stolen.)*	☐	☑
Feedback tells users that their action has been recognised.	☑	☐

D Exam-style questions

35 marks

1 A council is designing a self-service kiosk for paying parking fines. State four qualities its user interface should have. [4]

> Any four from: simple / clear, easy to learn (1) / intuitive and consistent (1) / responsive, giving feedback on status (1) / forgiving – mistakes easily reversed, help available (1) / attractive layout, typography and colour (1) / accessible (1)

2 State what each letter of WIMP stands for. [2]

> Windows / icons / menus / pointer: all four correct (2); two or three correct (1).

3 Give two advantages of a command line interface for an experienced network technician at a further education college. [2]

> Any two from: fast to use when the commands are known (1) / commands can be scripted to automate repetitive tasks (1) / needs little processing power or memory, so works on low-specification or remote devices (1) / gives precise, powerful control via switches/parameters (1)

4 Explain two disadvantages of a graphical user interface compared with a command line interface. [4]

> Needs more memory (RAM) (1) because graphics and several windows must be held in memory (1) / needs a more powerful processor and better display (1) to draw and update the graphics (1) / uses more storage space (1) / takes longer for programmers to develop (1) / repetitive tasks slower by mouse than by script (1). Two disadvantages, each identified (1) and explained (1).

5 A Derry~Londonderry theatre is designing an online booking system using a forms dialogue interface. Describe two advantages and one disadvantage of this type of interface for its customers. [3]

> Advantages, any two: easy to use, no commands to learn (1) / suitable for beginners as laid out logically (1) / data validated as entered, e.g. date of performance must be a valid date (1) / drop-downs limit choices so fewer errors (1). Disadvantage, any one: limited to options provided (1) / poorly designed form slow to complete (1) / tedious for frequent users (1)

6 Describe how a smartphone user could input text, sound and video. [3]

> Text: on-screen keyboard with predictive text/swipe typing, or speech-to-text dictation (1) / Sound: built-in microphone, e.g. voice notes, voice assistant (1) / Video: built-in front/rear camera records video, compressed on the device (1)

7 Explain what is meant by consistency in user interface design and why it is important. [2]

> The same layout, colours, terminology and controls are used throughout (1) / so users can predict how the interface works and transfer what they have learned, making it quicker to learn and reducing errors (1)

8 Describe three features that would make a council website accessible to a user with a visual impairment. [3]

> Any three from: compatible with screen-reader software (1) / alternative text for images (1) / text that can be resized without breaking the layout (1) / high colour contrast and colour not used as the only cue (1) / full keyboard navigation (1) / audio versions of key information (1)

9 Describe two disadvantages of voice interfaces. [4]

> Any two described, from: background noise interferes with recognition (1) causing wrong commands (1) / strong accents or speech impediments may not be understood (1) so some users are excluded (1) / homophones such as there/their confused (1) leading to wrong text (1) / may need training to a user's voice (1) which takes time (1) / privacy – others can overhear (1) / not suitable in quiet places such as a waiting room (1)

10 A GP surgery in Omagh is introducing a check-in system for patients of all ages. Discuss the suitability of a touch-screen forms or menu interface, a voice interface and a biometric interface for this system, and recommend one. [8]

> Indicative content: Touch-screen forms/menu – simple guided steps (enter date of birth, select appointment), large buttons, can be validated, quick, familiar from phones; but hygiene concerns, difficult for some older or visually impaired patients, small targets, needs accessibility features. Voice – hands-free, helps patients with limited sight or dexterity; but waiting room noise, accents, privacy as others overhear personal details, may misrecognise names. Biometric (fingerprint/face) – very fast, no need to remember details, identifies patient uniquely; but expensive, patients must enrol, privacy/GDPR concerns over special category data, some patients reluctant, fingerprint readers have hygiene issues. Recommendation: e.g. touch-screen with large text, high contrast and an option for reception staff assistance. Levels: 7–8 balanced discussion of all three with applied advantages and disadvantages and a justified recommendation; 4–6 discusses at least two with some application; 1–3 lists points with little context.

E Quick quiz

Circle one answer · 9 marks

1 Why might an experienced programmer prefer a command line interface to a GUI? [1]

A It has more colourful icons
B It is easier for beginners
C Knowing the commands, they can work faster and automate tasks with scripts ✓
D It needs a more powerful processor

2 What does WIMP stand for? [1]

A Windows, icons, menus, pointer ✓
B Web, internet, mail, protocol
C Wireless, input, memory, processor
D Windows, input, mouse, printer

3 Which is an advantage of a forms dialogue interface? [1]

A It is the fastest interface for experts
B Data can be validated as it is entered ✓

C It offers unlimited options
D It needs no screen

4 Which HCI principle does 'recognition rather than recall' support? [1]

A Feedback
B Consistency
C Error recovery
D Minimising memory load ✓

5 Which feature best helps a deaf user watching a training video? [1]

A Alt text
B High contrast
C Captions ✓
D Voice control

6 Which is a drawback of a GUI? [1]

A Hard for beginners
B Requires memorising syntax
C Cannot display images
D Needs more memory and processing power ✓

7 A spinning wheel appears while a file uploads. Which quality of a good interface does this show? [1]

A Responsive feedback ✓
B Biometric security
C Consistency of colour
D Forgiveness

8 Which input method lets a smartphone user enter text without typing? [1]

A Webcam
B Speech-to-text dictation ✓
C Trackpad
D Scanner

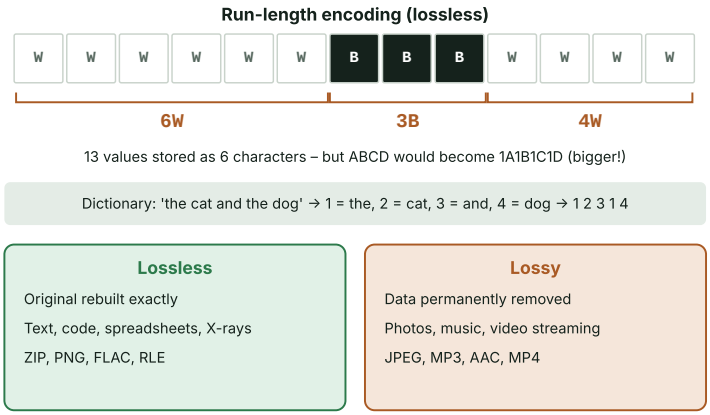
9 Which interface is most suitable for a surgeon who must keep their hands sterile? [1]

A Command line
B Forms dialogue
C Voice ✓
D Touch screen

Data compression and file formats

FD13 ANSWERS

I can: Explain why data is compressed and compare lossy and lossless compression · Apply run-length encoding and dictionary-based compression to simple examples and calculate the saving · Calculate the size of a bitmap image and a compression ratio · Choose and justify a file format and compression method for a given file and purpose



Use this diagram to help you: Run-length encoding of a row of pixels into 6W 3B 4W, and a comparison of lossless and lossy compression with example formats

A Key terms

Write the letter of the matching definition · 6 marks

C **Data compression**

E **Lossless compression**

F **Lossy compression**

D **Run-length encoding (RLE)**

A **Dictionary-based compression**

B **Compression ratio**

A A lossless compression method that stores repeated patterns once in a dictionary and replaces each occurrence with a short reference (token).

B The original file size divided by the compressed file size.

C Encoding data using fewer bits than the original representation, to reduce storage space and transmission time.

D A lossless compression method that replaces a run of identical values with a single value and a count.

E Compression that reduces file size without losing any data, so the original can be reconstructed exactly.

F Compression that permanently removes data judged less important, giving smaller files that cannot be restored exactly.

B Fill the gaps

Use the word bank · 9 marks

token | Run-length encoding | colour depth | storage | JPEG | Lossless | zipping | lossy | GIF

Compression reduces file size so that files need less **storage** space and take less time to transmit. **Lossless** compression allows the original file to be reconstructed exactly, while **lossy** compression permanently removes data, so it suits photographs and music. **Run-length encoding** replaces a run of identical values with a count and the value. Dictionary-based compression replaces repeated words with a short **token**. Combining several files into one compressed archive is

called **zipping**. A **GIF** image uses a palette of up to 256 colours, while a **JPEG** is best for photographs. The number of bits used for each pixel is the **colour depth**.

C True or false?

Tick one box · 9 marks

	True	False
RLE is a lossless compression method.	☑	☐
Lossy compression is suitable for program code. *(Any lost data would break the program; code needs lossless compression.)*	☐	☑
RLE can make some data larger.	☑	☐
MP3 removes sounds that humans are unlikely to hear.	☑	☐
Zipping a JPEG usually makes it much smaller. *(JPEG is already compressed, so zipping achieves little further saving.)*	☐	☑
A GIF image can show up to 256 colours.	☑	☐
WAV files are normally stored with lossy compression. *(WAV normally stores uncompressed audio, which is why the files are large.)*	☐	☑
Doubling the colour depth of a bitmap doubles its uncompressed file size.	☑	☐
PNG is a lossy image format. *(PNG uses lossless compression.)*	☐	☑

D Exam-style questions

33 marks

1 State two reasons for compressing files before uploading them to a school website. [2]

> To reduce the file size so it takes less storage space on the web server (1) / to reduce upload/download (transmission) time so pages load faster (1)

2 Explain the difference between lossy and lossless compression. [2]

> Lossy permanently removes some data, so the original cannot be restored exactly (1) / lossless removes redundancy without losing any data, so the original can be reconstructed exactly (1)

3 A bitmap image measures 3 inches by 4 inches, has a resolution of 150 dpi and a colour depth of 24 bits. Calculate the file size in bytes. Show your working. [3]

> Pixels: (3 × 150) × (4 × 150) = 450 × 600 = 270 000 pixels (1) / 24 bits = 3 bytes per pixel (1) / 270 000 × 3 = 810 000 bytes (≈ 791 KB) (1)

4 Use run-length encoding to compress the row of pixels RRRRRGGGBBBBBBB, and calculate the compression ratio. [3]

> 5R3G7B (2: 1 mark for two runs correct, 2 for all three) / 15 ÷ 6 = 2.5, ratio 2.5 : 1 (1)

5 Describe how dictionary-based compression reduces the size of a text file. [3]

> Repeated words/sequences are identified and each is stored once in a dictionary (1) / each occurrence in the text is replaced by a short token/index referring to the dictionary entry (1) / tokens take less space than the words, and the dictionary is stored with/rebuilt from the file so the text is restored exactly (lossless) (1)

6 A designer in Coleraine needs to email a large BMP logo to a printing company. Explain why zipping the file would be preferable to compressing it with a lossy algorithm. [2]

> Zipping is lossless, so the logo is reconstructed exactly with no loss of quality (1) / a lossy algorithm would

permanently lose detail, which could show when the logo is printed at high resolution (1)

7 Explain why GIF is suitable for a sports club's crest but not for a photograph of the team. [3]

GIF uses a palette of at most 256 colours (1) / a crest has few flat colours, so it is reproduced exactly with lossless compression and sharp edges, giving a small file (1) / a photograph contains thousands of shades, so reducing it to 256 colours causes banding/poor quality; JPEG would be better (1)

8 One minute of uncompressed stereo audio is stored as a WAV file at 44 100 samples per second with 16-bit samples. Calculate the file size in bytes. [3]

44 100 × 16 × 2 = 1 411 200 bits per second (1) / × 60 = 84 672 000 bits (1) / ÷ 8 = 10 584 000 bytes (≈ 10.1 MB) (1)

9 A music streaming service uses lossy compression. Explain how lossy compression reduces the size of audio files and why this is acceptable to users. [4]

Uses perceptual coding / psychoacoustics (1) / removes frequencies outside the range of human hearing (1) / removes quieter sounds masked by louder sounds at the same time (1) / acceptable because most listeners cannot hear the difference, and the much smaller files stream quickly using less mobile data (1)

10 A local newspaper in Ballymena is moving its archive of articles, photographs, audio interviews and video reports online. Evaluate the file formats and compression methods it should use for each type of content. [8]

Indicative content: Articles – TXT/HTML or a lossless format; text must be exact; text compresses well losslessly (ZIP/server compression); TXT is universally readable. Photographs – JPEG for archive photos on web pages, small files and fast loading, quality loss rarely noticeable; keep lossless master copies (e.g. TIFF/BMP/PNG) for reprinting; GIF unsuitable for photos due to 256 colours; GIF/PNG suit logos and graphics. Audio interviews – WAV for master recordings during editing (uncompressed, ~10 MB per minute); MP3/AAC for online playback (about 1 MB per minute at 128 kbps), quality loss acceptable for speech. Video reports – MPEG-4 (MP4), lossy, essential for streaming; encode once, watched many times; original footage kept separately. Zipping for transferring batches of files to the web host. Trade-offs: storage cost vs quality, loading speed for readers on mobile, permanent loss with lossy formats. Levels: 7–8 evaluates formats for all four content types with justified trade-offs and a conclusion; 4–6 suitable formats for most types with some justification; 1–3 names formats with little justification.

E Quick quiz

Circle one answer · 9 marks

1 A 6 MB bitmap photo is saved as a 0.5 MB JPEG. What is the compression ratio? [1]

A 3 : 1
B 6 : 1
C 12 : 1 ✓
D 1 : 12

2 Which file should use lossless compression? [1]

A A holiday photo
B A C# program ✓
C A streamed film
D A podcast

3 What is AAAABBBCCD in RLE? [1]

A 4A3B2C1D ✓
B A4B3C2D1
C 4A3B2CD
D ABCD4321

4 Which format uses lossy compression? [1]

A ZIP
B GIF
C BMP
D JPEG ✓

5 How many colours can 24-bit colour represent? [1]

A 24
B 256
C 65 536
D 16 777 216 ✓

6 Which data compresses best with RLE? [1]

A A photograph of a forest
B Random numbers
C A simple black-and-white icon ✓
D Encrypted data

7 Why does zipping an MP4 file save very little space? [1]

A MP4 files cannot be zipped
B The MP4 is already compressed ✓
C Zipping is lossy
D Zipping only works on text

8 Which format is best for a logo with five flat colours and sharp edges? [1]

A GIF ✓
B WAV
C JPEG
D TXT

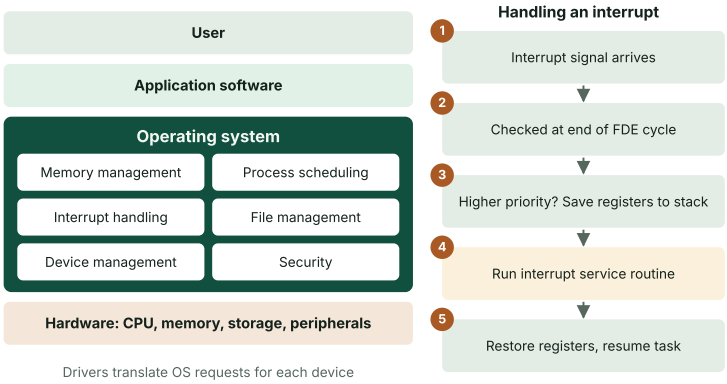
9 Why is MPEG well suited to online video services? [1]

A Video is encoded once but decoded and watched many times ✓
B It is lossless
C It stores every frame in full
D It only stores audio

Operating systems and utility programs

FD14 ANSWERS

I can: Explain what an operating system is for and describe the job of each of its main parts · Distinguish single user, multi-user, multiprogramming, multitasking, multiprocessing and multithreading systems · Explain how the OS manages memory, schedules processes and handles interrupts, and trace a round-robin schedule · Explain why utility programs are needed and describe compression, backup, archiving and defragmentation tools



Use this diagram to help you: The operating system sits between applications and hardware and provides memory management, scheduling, interrupt handling, file and device management and security; an interrupt saves registers, runs the service routine and restores them

A Key terms

Write the letter of the matching definition · 6 marks

C Operating system

A Kernel

F Device driver

B Multi-user operating system

D Time-slicing

E Multiprogramming

A The core of the operating system, which loads and unloads programs in memory and allocates memory, storage and processor time.

B An operating system that lets many users use one computer at the same time, sharing its resources by time-slicing while keeping each user's data private.

C System software that manages a computer's hardware and software resources and sits between the hardware, the application software and the user so that they can work together.

D Sharing processor time by giving each task or user a short, fixed period of CPU time in turn.

E Holding several programs in main memory at once and switching the CPU between them, although only one executes at any instant.

F Software that allows the operating system to communicate with and control a specific hardware device.

B Fill the gaps

Use the word bank · 9 marks

kernel | multiprocessing | defragmenter | operating system | device driver | multithreading | time-slicing | interrupt | Paging

The **operating system** manages the computer's resources and is the interface between hardware, applications and the user. Its core, the **kernel**, loads programs into memory and shares out processor time. A **device driver** lets the OS communicate with a particular piece of hardware. A multi-user OS uses **time-slicing** so that each user appears to have the computer to themselves. In **multiprocessing** the work is divided between several processors so it runs truly in parallel, whereas **multithreading** runs several paths of execution inside one process. **Paging** splits memory into fixed-size blocks. When an **interrupt** is serviced, the register contents are saved on the stack. A disk **defragmenter** makes each file's clusters contiguous so it can be read more quickly.

C True or false?

Tick one box · 9 marks

	True	False
The operating system is loaded before any application software when the computer starts.	☑	☐
A multi-user operating system gives each user a dedicated processor of their own. *(Users share the processor; the OS uses time-slicing, so each user seems to have the computer to themselves.)*	☐	☑
In a multiprogramming system several programs can be in main memory but only one is executed at any instant.	☑	☐
Multiprocessing means one processor switching rapidly between programs. *(Multiprocessing uses two or more processors or cores so that work genuinely runs in parallel; rapid switching on one processor is multiprogramming or multitasking.)*	☐	☑
Threads within the same process share memory and executable code.	☑	☐
Pages are always the same size, whereas segments vary in size.	☑	☐
The processor checks for interrupts at the end of each fetch–decode–execute cycle.	☑	☐
Deleting a file immediately erases its data from a hard disk. *(Deleting only marks the file's clusters as free; the data remains until those clusters are overwritten.)*	☐	☑
Solid state drives should be defragmented regularly to speed them up. *(SSDs have no moving read head, so defragmenting gives no benefit and adds wear; the OS uses TRIM instead.)*	☐	☑

D Exam-style questions

38 marks

1 State two purposes of an operating system. [2]

> Any two from: manages the computer's hardware resources (e.g. memory, processor, devices) (1) / provides an interface between the user and the hardware (1) / provides a platform/environment for application software to run and access the hardware (1) / manages files and security (1)

2 Describe the role of the kernel in an operating system. [2]

> Any two from: core of the OS (1) / loads programs into memory and unloads them when finished (1) / allocates resources such as memory and secondary storage (1) / schedules use of the processor (1)

3 A health trust's patient records system runs on a multi-user operating system used by hundreds of staff at the same time. Describe how the operating system makes sure that each [4]

user gets fair use of the system and that their data is kept private.

> Any four from: time-slicing shares processor time between users' tasks (1) / so each user seems to have the computer to themselves (1) / memory and storage are allocated so each user has sufficient resources (1) / data requests are queued/managed (1) / each user has a user account and password to authenticate (1) / access rights restrict each user to their own files/records (1) / one user's files are kept private from others (1)

4 Explain the difference between multiprogramming and multiprocessing. [4]

> Multiprogramming: several programs are resident in main memory (1) / only one executes at any one time; the OS switches when a program waits e.g. for I/O (1) / this gives the illusion of parallel processing / keeps CPU busy (1). Multiprocessing: uses more than one processor/core (1) / tasks are divided between processors so they run truly in parallel (1) / main benefit is speed (1). Max 4, at least one point on each.

5 Explain what is meant by multithreading, using a web server as an example. [3]

> A single process can have several threads (paths of execution) running concurrently (1) / threads share the process's memory and code (1) / e.g. a web server creates a thread for each visitor's request so many requests are handled without loading extra copies of the server program (1)

6 Processes P (needs 3 ms), Q (needs 5 ms) and R (needs 1 ms) arrive at time 0 in the order P, Q, R. The scheduler uses round robin with a 2 ms time slice. Show the order in which the processes run and state the time at which each one finishes. [4]

> Order: P 0–2, Q 2–4, R 4–5, P 5–6, Q 6–8, Q 8–9 (1 for first three slices correct, 1 for remaining order) / R finishes at 5 ms (1) / P finishes at 6 ms (1) / Q finishes at 9 ms (1). Max 4.

7 Describe what happens when the processor receives an interrupt from a printer that has run out of paper. [4]

> Any four from: the printer sends an interrupt signal to the processor (1) / the processor checks for interrupts at the end of the current fetch–decode–execute cycle (1) / the priority of the interrupt is compared with the current task (1) / the contents of the registers are saved on the stack (1) / the interrupt service routine for the printer is run, e.g. displaying an 'out of paper' message (1) / the register contents are restored and the original task resumes (1)

8 Describe two differences between paging and segmentation. [4]

> Any two differences, 2 marks each: pages are fixed size (1) whereas segments are variable size (1) / paging divides memory physically (1) whereas segments follow the logical structure of the program, e.g. a subroutine (1) / paging uses a page table mapping pages to frames (1) whereas segmentation uses a segment table with base address and length (1) / paging can waste space in the last page (internal fragmentation) (1) whereas segmentation leaves gaps between segments (external fragmentation) (1)

9 A solicitors' office in Omagh must keep closed client files for several years. Explain how a file archiving program would help the office. [3]

> Any three from: combines many client files into a single archive file / set of files (1) / usually compresses them so they take less storage space (1) / closed files can be moved off the main system to cheaper long-term storage, freeing space (1) / metadata such as file names, timestamps and access rights is stored so files can be rebuilt exactly when needed (1) / access to the archive can be restricted, protecting confidential client data (1) / retention period can be set so files are kept as long as required (1)

10 The network manager of a grammar school in Newry looks after older desktop PCs with magnetic hard disks and new laptops with solid state drives. Discuss how utility programs, including data compression, backup, archiving and disk defragmentation, can help the manager keep the school's computers efficient and its data safe. [8]

> Indicative content: compression reduces file sizes, saving storage on shared drives and speeding transfer across the network or to the cloud; backup utility can be scheduled overnight to make exact copies of pupil work and the MIS database, stored off-site/cloud, allowing restoration after hardware failure, accidental deletion or ransomware; full/incremental backups; archiving combines and compresses old data (e.g. leavers' folders), stores metadata and access rights, frees space and keeps data for required periods; defragmenter rearranges

clusters on magnetic hard disks so files are contiguous, reducing head movement and access time, best scheduled out of hours as it can take hours; must not be run on SSD laptops (no moving parts, adds wear; TRIM used instead); utilities often built into the OS and can be automated; some evaluation, e.g. backups must be tested, archives need the software to unarchive. Levels: 7–8 all four utilities described accurately and applied to both device types with a justified overall view; 4–6 three or four utilities described with some application to the school; 1–3 limited, generic description of one or two utilities.

E Quick quiz

Circle one answer · 9 marks

1 A bank's mainframe lets thousands of staff work at once, each appearing to have the computer to themselves. Which technique makes this possible? [1]

A Defragmentation
B Time-slicing ✓
C Data compression
D Segmentation

2 What happens to the contents of the processor's registers when a higher-priority interrupt is serviced? [1]

A They are deleted
B They are sent to the printer
C They are written to the hard disk as a file
D They are saved on the stack and restored afterwards ✓

3 Which term describes several threads running within a single process and sharing its memory? [1]

A Multithreading ✓
B Multiprocessing
C Multiprogramming
D Spooling

4 Which part of the operating system loads programs into memory and allocates processor time? [1]

A User interface
B Hardware interface
C Kernel ✓
D Device driver

5 Round robin, 2 ms slices: X needs 3 ms and Y needs 2 ms, X first. When does X finish? [1]

A 2 ms
B 3 ms
C 4 ms
D 5 ms ✓

6 What is the main benefit of a multiprocessing system? [1]

A Speed through true parallel processing ✓
B Smaller file sizes
C Fewer interrupts
D No need for drivers

7 Which memory management technique divides a program into variable-sized logical units? [1]

A Paging
B Segmentation ✓
C Spooling
D Caching

8 Why does a disk defragmenter speed up file access on a magnetic hard disk? [1]

A It compresses every file
B It deletes unused programs
C It places each file's clusters next to each other ✓
D It adds more RAM

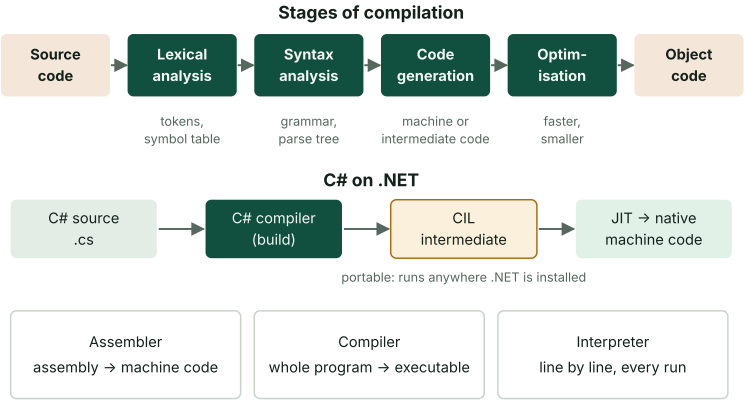
9 Which utility combines many files into one file, storing metadata so each can be rebuilt later? [1]

A Disk defragmenter
B Device driver
C Antivirus
D File archiving program ✓

Translators, application software and processing systems

FD15 ANSWERS

I can: Compare assemblers, compilers and interpreters and put the stages of compilation in order · Explain the differences between generic and special purpose software and weigh up custom-built against off-the-shelf software · Evaluate proprietary and open source software and the ways software is delivered to different platforms · Choose between batch and real-time processing for an application and justify the choice · Identify the input, processing and output of ATMs, POS terminals and smartphones



Use this diagram to help you: The stages of compilation from source code to object code, and how C# is compiled to Common Intermediate Language and then just-in-time compiled to machine code

A Key terms

Write the letter of the matching definition · 6 marks

D Compiler

A Interpreter

F Assembler

C Intermediate code

B Lexical analysis

E Application software

A A translator that translates and executes a high-level program one statement at a time, without producing an executable file.

B The first stage of compilation, in which comments and white space are removed, the source code is converted into tokens and a symbol table is built.

C Processor-independent code, such as Java bytecode or C# CIL, produced by a compiler and executed by a virtual machine or JIT compiler.

D A translator that converts a whole high-level program into object code before it is executed, producing an executable file.

E Programs designed to let the end user carry out specific tasks, such as word processing or booking tickets, accessing the hardware through the operating system.

F A translator that converts assembly language into machine code.

B Fill the gaps

Use the word bank · 10 marks

Open source | EFTPOS | lexical analysis | interpreter | real-time processing | proprietary | generic software | compiler | custom-built | batch processing

A **compiler** translates a whole high-level program before it runs, whereas an **interpreter** translates and executes one statement at a time. The first stage of compilation, **lexical analysis**, splits the source code into tokens. A word processor is an example of **generic software** because it can be used for many tasks without modification. Software written for one specific client is called **custom-built** or bespoke software. **Open source** software is distributed with its source code, which users may change, while **proprietary** software keeps its code secret. In **batch processing** transactions are collected and processed together at a set time, but an ATM withdrawal needs **real-time processing** so the balance is updated immediately. Paying in a shop by debit card uses **EFTPOS**.

C True or false?

Tick one box · 9 marks

	True	False
An interpreter produces an executable file that can be run without it. *(An interpreter translates and runs the code each time; no executable is produced, so the interpreter is needed whenever the program runs.)*	☐	☑
Syntax analysis checks that the tokens follow the grammar rules of the language.	☑	☐
Off-the-shelf software is usually cheaper than custom-built software because its development cost is shared among many buyers.	☑	☐
Freeware is the same as open source software. *(Freeware is proprietary software given away at no charge; its source code is not available and it cannot be modified.)*	☐	☑
A web application can be used on different operating systems as long as a suitable browser and internet connection are available.	☑	☐
In batch processing the master file is always completely up to date. *(The master file is only updated when the batch runs, so between runs it is out of date; real-time processing keeps it current.)*	☐	☑
Control totals are used in batch processing to check that no transactions have been lost or added.	☑	☐
A barcode scanner at a supermarket checkout is an output device. *(It is an input device: it reads the product code, which is then used to look up the price.)*	☐	☑
Java bytecode can run on any platform that has a Java Virtual Machine.	☑	☐

D Exam-style questions

41 marks

1 Describe two differences between a compiler and an interpreter. [4]

> Any two differences, 2 marks each: a compiler translates the whole program before it runs (1) whereas an interpreter translates and executes one line/statement at a time (1) / a compiler produces an executable/object code file (1) whereas an interpreter produces no executable so must translate every time it runs (1) / a compiler reports all errors after compilation (1) whereas an interpreter stops at the first error (1) / compiled programs run faster (1) because interpreted code is translated at run time (1) / compiled code can be distributed without the source code (1) whereas interpreted programs need the source code and interpreter (1)

2 Describe what happens during the lexical analysis and syntax analysis stages of compilation. [4]

Lexical analysis, any two from: white space and comments removed (1) / code split into tokens such as keywords, identifiers and operators (1) / symbol table of identifiers built (1). Syntax analysis, any two from: tokens checked against the grammar rules of the language (1) / parse/syntax tree built (1) / syntax errors reported (1)

3 A dental practice in Bangor uses a spreadsheet for its accounts and a dedicated package for appointments and patient records. Using these two programs, explain the difference between generic and special purpose software. [4]

Generic software can be used for many different tasks without modification (1), e.g. the spreadsheet could also be used for stock lists or rotas (1) / special purpose software is designed for one specific task or business need (1), e.g. the appointments package only manages dental bookings and patient records (1)

4 A Derry~Londonderry engineering firm is deciding whether to commission custom-built software to schedule its machine maintenance. Give two benefits and two drawbacks of this choice for the firm. [4]

Advantages, any two from: does exactly what the firm needs / optimised for its processes (1) / no unnecessary features taking up storage (1) / developers available to fix errors during early use (1) / may give a competitive advantage (1). Disadvantages, any two from: expensive to develop (1) / takes a long time to develop (1) / may contain undiscovered bugs triggered by a specific event (1) / little online support from blogs/forums (1)

5 A Belfast accountancy practice can either download and install its accounts package on each PC or use the same package through a web browser. Explain two benefits of using the web-based version. [4]

Any two benefits explained, 2 marks each: no installation needed on each PC (1) so less IT work and works on any device with a browser (1) / updates are applied centrally on the server (1) so all staff always use the latest version (1) / minimal processing power/storage needed on the client (1) so older PCs can be used (1) / data stored on the publisher's servers (1) so staff can access it from home or other offices (1)

6 Explain what is meant by open source software. [2]

Source code is distributed with the software / freely available (1) / users may modify it and redistribute it under the licence, usually free of charge (1)

7 State two differences between freeware and open source software. [2]

Freeware's source code is not available whereas open source's is (1) / freeware cannot be modified/redistributed as it is proprietary whereas open source can be changed and shared (1)

8 A water company sends quarterly bills to its business customers. Describe the main features of batch processing and explain why it is suitable for producing these bills. [4]

Features, any three from: meter readings/transactions collected over a period into a transaction file (1) / processed together as one batch at a set time (1) / without human intervention once set up (1) / usually run off-peak e.g. overnight (1) / control/batch totals used to validate the batch (1). Suitability (1): bills only needed once a quarter, so there is no need for immediate processing and the same calculation is applied to every customer efficiently. Max 4.

9 A customer withdraws £50 from an ATM. Identify one input, one processing activity and one output involved, and explain why the ATM system must use real-time processing. [5]

Input (1): e.g. card inserted / PIN entered / amount selected. Processing (1): e.g. PIN validated / account retrieved / balance checked / £50 deducted from balance. Output (1): e.g. cash dispensed / receipt printed / screen message. Real-time (2): the balance must be checked and updated immediately (1) so the customer cannot withdraw more than is in the account / overspend at another ATM or shop (1)

10 A chain of three bakeries in Co. Down wants new software to run its point of sale terminals and stock control. The owner's nephew is a software developer. Discuss the options available to the bakery, including off-the-shelf or custom-built software and proprietary or open source licences, and make a justified recommendation. [8]

> Indicative content: off-the-shelf POS/stock packages are available immediately, cheaper, widely tested, with support and training, but may need configuration and may not fit bakery-specific needs (e.g. batch production, wastage); custom-built by the nephew would fit exactly and could be cheaper than a software house, but takes longer, depends on one person for support and maintenance (risk if he leaves), may contain bugs, little online help; proprietary package gives vendor support and updates but costs licence fees and creates lock-in; open source POS software is free and could be adapted by the nephew because he can modify the source code, but may be less polished, documented or supported; web-based delivery would let the three shops share stock data in real time; consider real-time sales with batch reordering overnight. Recommendation must be justified from the scenario (e.g. an off-the-shelf web-based system for reliability and support, or open source adapted by the nephew to save cost). Levels: 7–8 balanced discussion of both pairs of options applied to the bakery with a justified recommendation; 4–6 several relevant points with some application, recommendation weakly justified; 1–3 generic advantages or disadvantages with little reference to the scenario.

E Quick quiz

Circle one answer · 9 marks

1 A health trust's payroll is calculated once a month for every employee, overnight and without staff intervention. What type of processing is this? [1]

A Real-time processing
B Interactive processing
C Batch processing ✓
D Multithreading

2 Which is the defining feature of open source software? [1]

A Its source code is available and may be modified ✓
B It is always free and never sold
C It has no licence of any kind
D It only runs on Linux

3 Which stage of compilation removes comments and splits code into tokens? [1]

A Code generation
B Lexical analysis ✓
C Optimisation
D Syntax analysis

4 Which is an advantage of custom-built software over off-the-shelf software? [1]

A It is available immediately
B It is cheaper
C It has large online user forums
D It does exactly what the organisation needs ✓

5 Which application most needs real-time processing? [1]

A Printing quarterly gas bills
B Booking seats on a flight ✓
C Calculating annual tax returns
D Producing monthly bank statements

6 Which of these happens in the processing stage at an ATM? [1]

A Inserting the card
B Dispensing the cash
C Checking the amount against the account balance ✓
D Printing a receipt

7 Why is a web application described as cross-platform? [1]

A It runs in a browser on any operating system ✓
B It is written in assembly language
C It is installed separately for each device
D It is always open source

8 Which smartphone component lets it pay contactlessly at an EFTPOS terminal? [1]

A Gyroscope
B Barometer
C Fingerprint sensor
D NFC antenna ✓

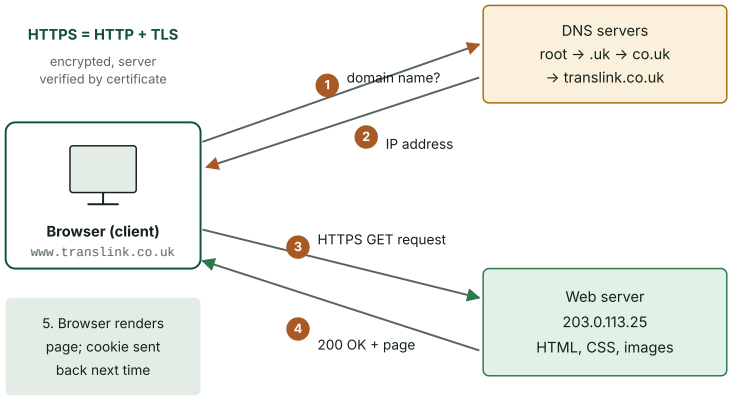
9 Which statement about off-the-shelf software is true? [1]

A It is written for one client
B It has usually been widely tested by many users ✓
C It never needs configuring
D It always fits a business exactly

How the internet and the web work

FD16 ANSWERS

I can: Distinguish the Internet from the World Wide Web and explain how websites are connected, hosted and uploaded · Convert an IPv4 address between binary and dotted decimal and break a domain name into its parts · Describe, in order, what happens when a URL is typed into a browser · Explain how cookies, sessions and caches overcome the limits of a stateless protocol · Describe the three parts of a search engine and the bodies that set web standards



Use this diagram to help you: Loading a web page: the browser asks DNS for the IP address of the domain name, then sends an HTTPS request to the web server, which returns the page

A Key terms

Write the letter of the matching definition · 6 marks

B Internet

A World Wide Web

F IP address

D Domain Name System (DNS)

C Top-level domain (TLD)

E URL

A A collection of linked web pages and resources, each with a unique address, accessed over the Internet using HTTP and browsers.

B A global network of interconnected computer networks that use TCP/IP so that any connected device can communicate with any other.

C The last part of a domain name, such as .uk, .ie or .com, which may indicate the country or type of organisation.

D A distributed, hierarchical system of servers that translates domain names into IP addresses.

E Uniform resource locator: the full address of a resource on the Web, made up of the protocol, domain name and path or file name.

F A unique numeric address assigned to a device on a network such as the Internet: 32 bits in IPv4 and 128 bits in IPv6.

B Fill the gaps

Use the word bank · 10 marks

IP address · cache · FTP · stateless · Domain Name System · top-level domain · Internet · web browser · World Wide Web · cookie

The **Internet** is a global network of networks that uses TCP/IP, while the **World Wide Web** is a service of linked pages reached over it. Each connected device has a unique **IP address**, which is 32

bits long in IPv4. In a domain name, the final label, such as .uk, is the **top-level domain**. The **Domain Name System** converts a domain name into an IP address. A **web browser** requests pages from a web server using HTTP, which is a **stateless** protocol, so sites use a **cookie** stored on the user's device or a session variable stored on the server to remember the user. Content that has already been downloaded can be reused from a **cache**, and website files are uploaded to a web host using **FTP**.

C True or false?

Tick one box · 10 marks

	True	False
The World Wide Web and the Internet are two names for the same thing. *(The Internet is the global network of networks; the Web is one service, a collection of linked pages, that runs over it.)*	☐	☑
An IPv4 address is made up of 32 bits.	☑	☐
IPv6 addresses are written as four decimal numbers separated by full stops. *(IPv6 addresses are written as eight groups of hexadecimal digits separated by colons; four decimal octets is the IPv4 format.)*	☐	☑
In www.bbc.co.uk the top-level domain is uk.	☑	☐
DNS translates IP addresses into the MAC addresses of network cards. *(DNS translates domain names into IP addresses.)*	☐	☑
HTTP is stateless, which means each request is independent of earlier requests.	☑	☐
Session variables are stored on the user's computer. *(Session variables are stored on the web server; the browser holds only a session ID, usually in a cookie.)*	☐	☑
A proxy cache can reduce the Internet traffic of a whole organisation.	☑	☐
The W3C is responsible for standards such as HTML and CSS.	☑	☐
FTP is only used for downloading files, never for uploading. *(FTP transfers files in both directions and is commonly used to upload website files to a web host; only anonymous FTP is limited to public downloads.)*	☐	☑

D Exam-style questions

42 marks

1 Explain the difference between the Internet and the World Wide Web. [2]

> The Internet is a global network of interconnected networks / the physical infrastructure and protocols (1) / the World Wide Web is a service running over the Internet: a collection of linked web pages accessed with a browser (1)

2 Explain the meaning of each of the terms web page, website and web server, and describe the relationship between them. [4]

> Web page: a single hypertext document viewed in a browser (1) / website: a set of related web pages with a common domain name (1) / web server: computer/software that stores web pages and delivers them when requested by a browser using HTTP (1) / relationship: a website is made up of web pages, stored on and delivered by a web server, and together all websites form the WWW (1)

3 The IPv4 address of a council's web server is written in binary as 11000011 00010110 00001010 01000001. Convert it to dotted decimal notation, showing your working. [3]

> 11000011 = 128 + 64 + 2 + 1 = 195 / 00010110 = 16 + 4 + 2 = 22 / 00001010 = 8 + 2 = 10 / 01000001 = 64 + 1 = 65 (1 for working shown for at least two octets, 1 for at least three octets correct) / final answer 195.22.10.65 (1)

4 Explain why IPv6 was introduced and state one difference between the format of IPv4 and IPv6 addresses. [3]

> IPv4 provides about 4.3 billion addresses (32 bits) (1) / growth in connected devices means there are not enough unique addresses (1) / difference, one from: IPv4 is 32 bits whereas IPv6 is 128 bits / IPv4 written as four decimal numbers separated by full stops whereas IPv6 is eight groups of hexadecimal separated by colons (1)

5 Identify the top-level domain and each of the sub-domains in the domain name www.qub.ac.uk. [3]

> Top-level domain: uk (1) / sub-domains: ac.uk (1) / qub.ac.uk and www.qub.ac.uk (www.qub.ac.uk is a sub-domain of qub.ac.uk, which is a sub-domain of ac.uk) (1)

6 A pupil types https://www.nimuseums.org.uk/visit into a browser. Describe the steps that take place before the page is displayed. [5]

> Any five in a logical order: browser checks its cache (1) / browser sends the domain name www.nimuseums.org.uk to a DNS resolver/server (1) / DNS servers are queried and the IP address of the web server is returned (1) / browser connects to the web server using the IP address and a secure TLS connection is set up because of https, with the server's certificate checked (1) / HTTP GET request for the /visit page is sent (1) / server returns a status code and the HTML (1) / browser requests associated CSS, images and scripts and renders the page (1)

7 A sports club in Ballymena adds new pages to its website. Name the protocol used to upload the pages to the web host and the one used when a member views them, explaining each choice. [4]

> Upload: FTP (1) because FTP transfers files between computers / allows authenticated upload to the web host with username and password (1). Viewing: HTTP (or HTTPS) (1) because HTTP is the protocol a browser uses to request and receive web pages from a web server (1)

8 HTTP is a stateless protocol. State two advantages of this and explain how an online shop can still remember the contents of a customer's basket. [4]

> Advantages: simple to implement (1) / efficient, using minimal network/server resources (1). Basket: a cookie containing a basket or session ID is stored by the browser (1) and sent back with each request so the server can link requests to the same basket, or the basket is held in a session variable on the server identified by that ID (1)

9 A local newspaper wants its articles to appear in search results. Describe the job of each of the three main parts of a search engine. [6]

> Web crawler: starts from seed pages and follows every link (1) / adds unvisited pages to its list until all pages visited or a limit is reached (1). Index: stores information about each page visited, e.g. text, images, metadata (1) / in a dictionary structure linking content to the page (1). Query engine: compares the user's search terms with the index (1) / returns matching pages ranked e.g. by PageRank so best matches are first (1)

10 A further education college in Enniskillen is concerned about slow web access and the privacy of student data. Discuss how caching, cookies, sessions, HTTPS and the work of standards bodies such as the W3C and ITU affect the speed, reliability and security of the college's use of the Web. [8]

> Indicative content: a proxy cache shares frequently used content between students, cutting bandwidth use and speeding up access; browser caches avoid re-downloading logos/menus; risk of stale content for frequently changing pages; cookies let sites remember logins and preferences but tracking cookies raise privacy issues and need consent under PECR/UK GDPR; session variables keep sensitive state on the server, more secure than cookies; HTTPS (TLS) encrypts data in transit and certificates authenticate servers, protecting student logins and personal data; standards bodies: W3C standards (HTML, CSS, accessibility) mean college pages and

systems work in any browser; ITU standards for transmission, encoding and certificates make communication reliable and interoperable; IETF protocols. Evaluation, e.g. caching improves speed but must be configured; security depends on users and settings as well as protocols. Levels: 7–8 all areas discussed accurately and applied to the college with balanced judgement; 4–6 most areas covered with some application; 1–3 limited, generic points on one or two areas.

E Quick quiz

Circle one answer · 9 marks

1 When you type a web address into a browser, what does the Domain Name System do? [1]

A Encrypts the page
B Stores the page's images
C Translates the domain name into the server's IP address ✓
D Checks the page for viruses

2 Why does HTTP need cookies or session variables for online banking? [1]

A HTTP cannot carry text
B HTTP is stateless, so the server would not otherwise know the user had logged in ✓
C HTTP only works on intranets
D HTTP deletes pages after viewing

3 What is 11000001 in denary, as one octet of an IPv4 address? [1]

A 192
B 129
C 65
D 193 ✓

4 Which protocol would a developer normally use to upload website files to a web host? [1]

A FTP ✓
B HTTP
C DNS
D TLS

5 How many bits are in an IPv6 address? [1]

A 32
B 128 ✓
C 64
D 256

6 Which search engine component follows links from page to page? [1]

A Index
B Query engine
C Web crawler ✓
D Cache

7 Which organisation agrees the standards for HTML and CSS? [1]

A W3C ✓
B ITU
C ISP
D DNS

8 In www.stran.ac.uk, what is stran.ac.uk? [1]

A The top-level domain
B The protocol
C The file name
D A sub-domain of ac.uk ✓

9 What is the main benefit of a browser cache? [1]

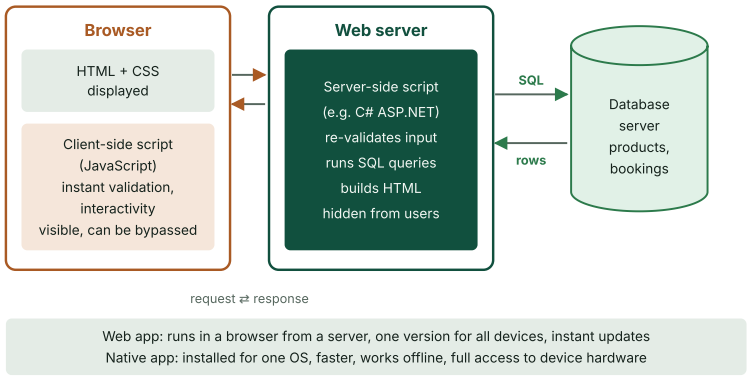
A Pages already fetched can be shown without downloading them again ✓
B It encrypts passwords
C It assigns IP addresses
D It blocks all cookies

AS 2: FUNDAMENTALS OF DIGITAL TECHNOLOGY · WEB APPLICATIONS

Web applications, intranets and web security

FD17 ANSWERS

I can: Decide whether a task belongs on the client or the server and explain how a dynamic page is built from a database · Weigh up web apps against native apps for a given purpose · Describe e-services, e-government and portals, and evaluate an intranet for an organisation · Explain how public and private keys, TLS/HTTPS, digital certificates and digital signatures keep data secure over the internet



Use this diagram to help you: A dynamic web page: client-side script runs in the browser, while the server-side script on the web server queries the database server and builds the HTML

A Key terms

Write the letter of the matching definition · 6 marks

B Dynamic web page

D Client-side scripting

F Server-side scripting

E Web application

A Native application

C SQL injection

A An application written for and installed on a specific operating system, with direct access to device hardware.

B A web page generated at the time of request, often from database data, so its content can change or be personalised.

C An attack in which SQL code is entered into an input field so that it is executed by the database.

D Code, usually JavaScript, that is downloaded with a web page and run by the user's browser.

E An application that runs in a web browser and is delivered from a server, processing user input and usually storing its data in a database on the server, without being installed on the device.

F Code that runs on the web server to process requests, access databases and generate pages before they are sent to the browser.

B Fill the gaps

Use the word bank · 10 marks

intranet | server-side scripting | client-side scripting | dynamic web page | private key | digital certificate | public key | e-service | digital signature | HTTPS

A **dynamic web page** is generated when it is requested, often from a database. Code that runs in the browser is **client-side scripting**, while database queries are handled by **server-side scripting**. Online

banking and shopping are examples of an **e-service** . A private network of web pages that only an organisation's staff can access is an **intranet** . In public key encryption, a message is encrypted with the recipient's **public key** and decrypted with their **private key** . Web pages sent over an encrypted TLS connection use **HTTPS** . A **digital certificate** issued by a certificate authority proves who owns a public key, and a **digital signature** proves that a document is genuine and unaltered.

C True or false?

Tick one box · 9 marks

	True	False
Client-side validation alone is enough to protect a database from malicious input. *(Client-side checks can be bypassed, so input must also be validated on the server, using parameterised queries to prevent SQL injection.)*	☐	☑
A single web app can run on devices with different operating systems.	☑	☐
E-government services remove the need for any paper or face-to-face alternatives. *(Governments must serve all citizens, so alternatives must remain for people without internet access, equipment or skills.)*	☐	☑
An intranet can only be accessed by authorised members of the organisation.	☑	☐
In public key encryption, a message for Aoife is encrypted with Aoife's public key.	☑	☐
A digital certificate contains the owner's private key. *(A certificate contains the owner's public key; the private key is kept secret by its owner and is never shared.)*	☐	☑
A digital signature is created by encrypting a hash of the document with the sender's private key.	☑	☐
TLS has replaced SSL as the protocol used to secure web connections.	☑	☐
Symmetric encryption uses a different key to decrypt from the one used to encrypt. *(Symmetric encryption uses the same secret key for both; asymmetric (public key) encryption uses a linked pair of keys.)*	☐	☑

D Exam-style questions

38 marks

1 Explain the difference between a static and a dynamic web page. [2]

> A static page is a fixed HTML file that is the same for every visitor until edited (1) / a dynamic page is generated when requested, e.g. by a server-side script using database data, so content can change or be personalised (1)

2 Describe one advantage and one disadvantage of client-side scripting. [4]

> Advantage (2), one from: runs in the browser so gives an immediate response (1) e.g. instant validation feedback (1) / reduces load on the server/network (1) because fewer requests are sent (1). Disadvantage (2), one from: code is visible/can be altered or disabled by the user (1) so cannot be relied on for security/validation (1) / behaviour may differ between browsers (1) so must be tested on many (1)

3 Explain why form data should be validated on the server even if it has already been validated in the browser. [2]

> Client-side validation can be bypassed, e.g. by disabling JavaScript or sending a request directly (1) / so the server must check data to protect the database, e.g. against SQL injection or invalid data (1)

4 A bus company's website lets customers see times for one route. Describe how a server-side script produces the page for the route a customer chooses. [4]

> Any four from: the customer's route choice is sent to the server, e.g. in the URL query string or a form (1) / server-side script receives and validates the route (1) / script runs a query on the database for that route's times (1) / database returns the matching records (1) / script builds an HTML page containing the times (1) / page is sent to the browser and displayed (1)

5 Describe two benefits to a high street shop in Lisburn of also offering its products as an e-service. [4]

> Any two benefits described, 2 marks each: access to a wider customer base (1) beyond customers who can visit the shop in Lisburn (1) / an additional channel for selling/contacting customers (1) so sales can be made 24/7 (1) / can offer a wider range of products than shelf space allows (1) / more data about customers' preferences (1) for targeted marketing / competitive advantage (1)

6 Explain what is meant by a portal, giving an example. [3]

> A website or page acting as a single starting point that links to many related services (1) / in a consistent manner / from one starting point, often after a single log-in, with groups of links that can be rearranged (1) / example, e.g. a government portal linking to departments' services or a school portal linking to email, timetable and homework (1)

7 Explain how public and private keys are used so that a patient's test results can be sent securely from a laboratory to a GP practice. [4]

> The GP practice has a key pair: a public key available to anyone and a private key kept secret (1) / the laboratory encrypts the results using the GP practice's public key (1) / the encrypted results are sent; if intercepted they cannot be read (1) / only the GP practice's private key can decrypt the results (1)

8 Describe how a digital signature allows a council to confirm that a planning document it has received has not been altered. [4]

> Sender creates a hash of the document (1) / encrypts the hash with its private key to form the signature and sends it with the document (1) / council decrypts the signature with the sender's public key (1) / council recalculates the hash of the received document and compares; if they match the document is unaltered / if not it has been changed (1)

9 State three items of information contained in a digital certificate. [3]

> Any three from: owner's name/organisation (1) / domain name of the website (1) / owner's public key (1) / serial number (1) / name of the issuing certificate authority and its signature (1) / expiry date / validity period (1)

10 A manufacturing company with sites in Craigavon and Coleraine is considering introducing an intranet for its 600 employees. Evaluate the use of an intranet by the company, including the security measures it should put in place. [8]

> Indicative content: advantages – policies, forms, health and safety guidance and manuals online in one up-to-date version for both sites; role-based access targets information to groups (shop floor, office, managers); messaging, forums and wikis improve communication between Craigavon and Coleraine; online leave requests, scheduling, surveys; reduced printing. Disadvantages – set-up and maintenance cost, staff to maintain content; personal data must comply with UK GDPR/DPA 2018; risk of unauthorised access through weak passwords, especially with remote access; dependence on availability; staff training. Security measures – strong passwords and multi-factor authentication; access rights by role; HTTPS/TLS for all intranet pages; VPN for remote access; firewall; encryption of stored sensitive data; backups; audit logs; staff training. Levels: 7–8 balanced evaluation applied to the company with appropriate security measures and a justified conclusion; 4–6 advantages and disadvantages with some application and some security measures; 1–3 limited, generic list of points.

E Quick quiz

Circle one answer · 9 marks

1 Ciara wants to send a confidential file to Declan using public key encryption. Which key should she use to encrypt it? [1]

A Her own private key
B Declan's private key
C Declan's public key ✓
D Her own public key

2 What does a digital signature prove about a document? [1]

A That it was sent over Wi-Fi
B Who sent it and that it has not been altered ✓
C That it contains no viruses
D That it is stored in the cloud

3 Which task should always be carried out by a server-side script? [1]

A Checking login details against a database ✓
B Highlighting an empty form field as the user types
C Animating a drop-down menu
D Showing a password-strength bar

4 Who issues and signs a digital certificate? [1]

A The user's ISP
B The web browser
C The W3C
D A certificate authority ✓

5 Which is an advantage of a native app over a web app? [1]

A Works on any device with a browser
B Full access to device hardware and works offline ✓
C Updated instantly for all users
D No app store needed

6 Which is a disadvantage of an intranet? [1]

A Staff see only the latest documents
B Information can be targeted to groups
C It costs a lot to set up and keep up to date ✓
D It improves internal communication

7 What is HTTPS? [1]

A HTTP sent over an encrypted TLS connection ✓
B A faster version of FTP
C A type of web browser
D A search engine protocol

8 Renewing a driving licence on a government website is an example of what? [1]

A An intranet
B A native app
C Symmetric encryption
D E-government ✓

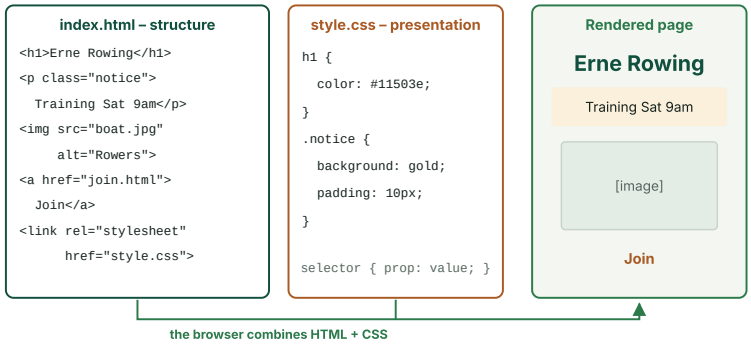
9 Why does TLS use public key encryption only to set up the connection? [1]

A Public key encryption cannot encrypt text
B Public key encryption is slow, so a faster symmetric session key encrypts the data ✓
C Browsers do not support symmetric encryption
D It makes certificates unnecessary

HTML and CSS

FD18 ANSWERS

I can: Read, write and correct HTML for a page with headings, paragraphs, lists, hyperlinks, images and a simple table or form · Explain the difference between empty and container tags and use attributes such as href, src, alt and type correctly · Write CSS rules using element, class and id selectors and specify colours by name, rgb() and hexadecimal · Work out how text will be displayed when inline, internal and inherited styles conflict · Explain why an external style sheet makes a multi-page website easier to maintain



One external style sheet styles every page: consistent, quick to change, cached by the browser

Use this diagram to help you: HTML provides structure and content, an external CSS style sheet provides presentation, and the browser combines them into the rendered page

A Key terms

Write the letter of the matching definition · 6 marks

D HTML

A Container tag

F Empty tag

E Attribute

B Anchor tag

C Alt text

A An HTML tag that is used as an opening and closing pair enclosing content, such as … or <a> … .

B The <a> tag, which creates a hyperlink; its href attribute gives the destination URL and the enclosed content is what the user clicks.

C The alt attribute of an image, giving a text description that is displayed if the image cannot load and is read aloud by screen readers.

D Hypertext Markup Language: the markup language that uses tags to define the structure and content of a web page, such as headings, paragraphs, lists, images and links.

E Extra information written inside an HTML opening tag as name="value", which refines the element, such as href, src or alt.

F An HTML tag that stands alone with no closing tag because it encloses no content, such as
, <img> or <link>.

B Fill the gaps

Use the word bank · 9 marks

precedence | empty | external | ol | selector | attribute | structure | anchor | class

HTML describes the **structure** and content of a web page, while CSS controls its presentation. A tag such as
 that has no closing tag is called an **empty** tag. Extra information inside an opening tag, such as href or alt, is called an **attribute**. A hyperlink is created with the **anchor** tag. A numbered list uses the **ol** tag, with each item inside li tags. In a CSS rule, the **selector** picks out the elements to be styled. A **class** selector begins with a full stop and can be used on many elements. A style defined closer to an element overrides one further away; this is called **precedence**. Linking every page to an **external** style sheet keeps the house style consistent across the site.

C True or false?

Tick one box · 9 marks

	True	False
The <img> tag is an empty tag because it has no closing tag.	☑	☐
Pressing Enter several times in the HTML source creates blank lines on the displayed page. *(Browsers collapse white space into a single space; use , or CSS to control spacing.)*	☐	☑
The alt attribute provides a text alternative for an image that screen readers can read aloud.	☑	☐
An id selector can be used to style many elements on the same page. *(An id must be unique on a page; use a class to style many elements.)*	☐	☑
The value #FF0000 in CSS represents pure red.	☑	☐
An internal style sheet automatically applies to every page on the website. *(An internal style sheet only affects the page whose head it is in; an external style sheet can be linked to every page.)*	☐	☑
An inline style on an element overrides a conflicting rule for that element in the page's style sheet.	☑	☐
The <head> section contains the text and images that the user sees. *(Visible content goes in the <body>; the head holds the title, metadata and links to style sheets.)*	☐	☑
A span inside a paragraph inherits the paragraph's font-family unless a different font is set for the span.	☑	☐

D Exam-style questions

40 marks

1 HTML elements are written using either container tags or empty tags. Describe what is meant by each type of tag and give one example of each. [4]

> Container tag: comes as an opening and closing pair (1) which encloses content (1), e.g. ... , <a> ... , <h1> ... </h1> (1) / Empty tag: stands alone with no closing tag/no content, e.g.
, <img>, <link> (1).

2 A walking club in the Mourne Mountains wants a page that shows the heading 'Upcoming walks', a sentence 'All walks start at 10am.', a numbered list of three walks (Slieve Donard, Slieve Binnian, Slieve Meelmore) and a link with the text 'Book a place' that opens booking.html. Write the HTML for the body of this page. [6]

> <h1>Upcoming walks</h1> (1) / All walks start at 10am. (1) / ... used for the numbered list (1) / three <li> ... items with the correct text (1) / (1) ... Book a place with correct

closing tag (1). Accept any heading level; accept ul only if numbering added by other means (not expected).

3 Write the HTML to display the image logo.png, stored in a folder called images, so that clicking the image takes the user to index.html. Include suitable alternative text. [3]

 (1) / <img src="images/logo.png" (1) alt="[suitable description, e.g. Club logo – home page]"> (1) . The img must be inside the anchor for full marks.

4 State the purpose of the alt attribute in an img tag and explain why it is important for some users. [2]

Provides alternative text describing the image (1) / which is read aloud by screen readers for visually impaired users, or shown if the image cannot load (1).

5 Write a CSS rule that makes every element with the class 'result' display in navy Arial text on a light grey (#EEEEEE) background. [3]

.result (1) { color: navy; font-family: Arial; (1) background-color: #EEEEEE; } (1). Allow named equivalents; one mark lost for missing full stop on the selector.

6 A designer wants to show text in pure blue. Give three different CSS values for the color property that would all produce this colour. [3]

Any three from: blue (1) / rgb(0, 0, 255) (1) / #0000FF (1) / #00F (1).

7 A page contains: <style> p { color: green; font-family: Verdana; } </style> and later Full time . (a) Describe how the word 'time' will be displayed, explaining your answer. (b) State how the word 'Full' will be displayed. [5]

(a) Orange (1) because the inline style is closer to the span and takes precedence over the inherited green (1); Verdana font (1) because font-family is inherited from the p rule (1). (b) Green Verdana text (1)

8 Explain why a web developer would write < rather than typing the < character when it is needed as text on a page. [2]

The < character has a special meaning in HTML as the start of a tag (1) so the browser could misinterpret it / the entity < tells the browser to display the symbol literally (1).

9 Compare inline styles with an internal style sheet in the head of a page. [4]

Inline: written in the style attribute of an individual element (1); affects only that element and must be repeated on every element (1) / Internal: written in a <style> block in the head (1); one rule applies to every matching element on that page, so it is easier to maintain than inline, but only for that page (1). Inline overrides internal when they conflict (1). Max 4.

10 Fermanagh and Omagh District Council's website has over 300 pages built by several staff. Some pages use inline styles and others have their own internal style sheets. Evaluate the benefits of moving the whole site to a single external style sheet. [8]

Indicative content: one CSS file linked to every page via <link>; a single change (e.g. new council branding colours) is applied to all 300 pages, reducing maintenance time and cost; removes inconsistencies between pages written by different staff, giving a consistent house style and professional image; HTML files are smaller and the CSS file is cached once, so pages load faster and use less bandwidth; content editors can focus on content while a designer controls appearance; alternative style sheets for mobile, print or high contrast can be added without changing content, improving accessibility. Drawbacks/evaluation: initial effort to remove inline styles and test every page; existing inline styles would still override the external file until removed; if the CSS file fails to load, all pages lose their styling; staff need training. Levels: 7–8 a balanced evaluation with several well-explained benefits applied to the council context and a justified conclusion; 4–6 several benefits explained with some reference to context; 1–3 benefits listed with little explanation.

E Quick quiz

Circle one answer · 9 marks

1 A web page needs a numbered list of opening times. Which tags should be used? [1]

A <ul> with <li> items
B <list> with <item> items
C <ol> with <li> items ✓
D <dl> with <dt> items

2 A style sheet says p { color: blue; } but one paragraph has style="color: red". What colour is that paragraph's text? [1]

A Red, because the inline style is closest to the element and takes precedence ✓
B Blue, because the style sheet always wins
C Purple, because the two colours are blended
D Black, because conflicting rules cancel out

3 Which tag correctly creates a link to the page results.html? [1]

A <link href="results.html">Results</link>
B Results ✓
C Results
D <href="results.html">Results</href>

4 Which CSS selector styles every element with class="notice"? [1]

A #notice
B notice
C *notice
D .notice ✓

5 Which of these is an empty tag? [1]

**A
 ✓**
B <p>
C
D <a>

6 Which CSS value gives the same colour as rgb(0, 128, 0)? [1]

A #001280
B #000080
C #008000 ✓
D #800000

7 Why is & used in HTML instead of typing &? [1]

A It makes the text bold
B The ampersand starts a character entity, so it must be escaped to display literally ✓
C Browsers cannot display the & symbol at all
D It is required by CSS

8 What is the main benefit of linking every page of a site to one external style sheet? [1]

A Pages no longer need any HTML
B Images are compressed automatically
C Form data is validated on the server
D A single change to the CSS file updates the appearance of every page ✓

9 Where is the <title> element placed and where is it shown? [1]

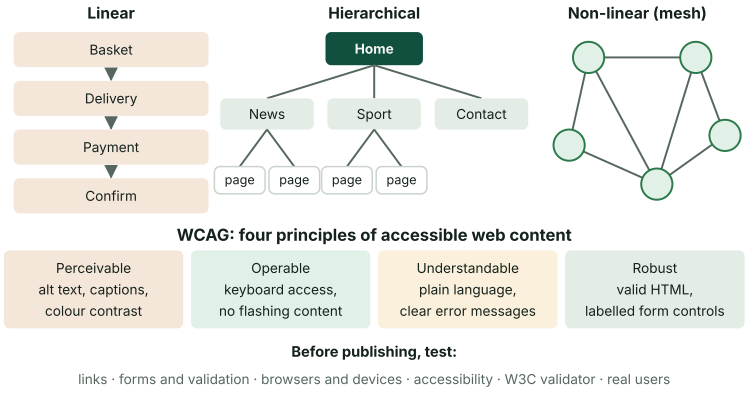
A In the head; on the browser tab ✓
B In the body; as the main heading
C In the body; at the bottom of the page
D In the CSS file; in the address bar

AS 2: FUNDAMENTALS OF DIGITAL TECHNOLOGY · WEBSITE DEVELOPMENT

Website design, client-side and server-side processing, testing and publishing

FD19 ANSWERS

I can: Plan a website using purpose, audience, a site map and storyboards, and justify a navigation structure · Apply usability and WCAG accessibility principles to a page design · Distinguish client-side from server-side processing using a real scenario such as an online booking · Explain why input is validated on both the client and the server · Describe the tests carried out before a website is published and the steps needed to publish it



Use this diagram to help you: Linear, hierarchical and non-linear navigation structures, and the four WCAG principles: perceivable, operable, understandable and robust

A Key terms

Write the letter of the matching definition · 6 marks

B Target audience

C Site map

F Storyboard

A Hierarchical structure

E Linear structure

D WCAG

A A navigation structure in which the home page links to main sections, which in turn link to their own sub-pages.

B The group of people a website is designed for, described by factors such as age, interests, abilities, accessibility needs and the devices they use.

C A diagram showing all the pages of a website and how they are linked.

D Web Content Accessibility Guidelines: the international standard for accessible web content, based on the principles perceivable, operable, understandable and robust.

E A navigation structure in which pages are visited in a fixed sequence using next and previous links.

F A sketch of the layout of each web page, showing the position of navigation, text, images and other elements, often with annotations.

B Fill the gaps

Use the word bank · 9 marks

linear | broken link | responsive | storyboard | hierarchical | server-side | robust | site map | client-side

A **site map** shows all the pages of a website and how they are linked, while a **storyboard** shows the layout of each page. In a **hierarchical** structure, the home page branches into sections and sub-pages, whereas an online checkout normally uses a **linear** structure. The four WCAG principles are perceivable, operable, understandable and **robust**. A layout that adapts to different screen sizes uses **responsive** design. Checking that required fields are filled in before data is sent is an example of **client-side** processing. Updating a booking database is an example of **server-side** processing. A link that leads to a missing page is called a **broken link**.

C True or false?

Tick one box · 9 marks

	True	False
A linear navigation structure suits an online checkout process.	☑	☐
WCAG stands for Website Colour and Graphics standard. *(WCAG stands for Web Content Accessibility Guidelines, published by the World Wide Web Consortium.)*	☐	☑
Alt text helps screen-reader users understand images.	☑	☐
Client-side validation alone is enough to keep a web application's database safe. *(Client-side checks can be bypassed, so the server must validate the data again before using or storing it.)*	☐	☑
Checking whether a username is already taken must be done on the server.	☑	☐
A static web page needs a database to work. *(A static page is a finished file that is simply retrieved and displayed; web applications use databases.)*	☐	☑
A storyboard shows the layout of individual pages.	☑	☐
Testing in one browser is enough to ensure compatibility. *(Browsers and devices can render pages differently, so several browsers and screen sizes must be tested.)*	☐	☑
Once a website is published, no further maintenance is needed. *(Content must be updated, broken links fixed, security updates applied and backups taken.)*	☐	☑

D Exam-style questions

40 marks

1 State two things a developer should find out about the target audience before designing a website. [2]

> Any two from: age range (1) / interests or purpose of their visit (1) / reading level or IT skills (1) / accessibility needs, e.g. visual impairment (1) / the devices they use, e.g. mobile phones (1) / language (1).

2 Describe what a site map shows and explain why one is produced before a website is built. [3]

> Shows every page in the website (1) and how the pages are linked / the navigation structure (1) / produced first so no page is missing or unreachable and the navigation can be agreed before coding begins (1).

3 A Belfast arts centre is designing its website. It has sections for shows, workshops and visiting, and a ticket-buying process. Describe a suitable navigation structure for the site, justifying your choice. [4]

> Hierarchical/mixed structure identified (1) / home page links to main sections: shows, workshops, visiting (1) / each section links to its own sub-pages, e.g. individual shows (1) / ticket buying uses a linear sequence (select

seats, details, payment, confirmation) because steps must be completed in order (1). Justification must relate to the context.

4 Describe three features that would make a hospital trust's website more accessible, stating who each feature would help. [6]

Three features, each with description (1) and who it helps (1), e.g.: alt text on images (1) read aloud by screen readers for blind/visually impaired users (1) / captions or transcripts on videos (1) for deaf or hard-of-hearing users (1) / high colour contrast and resizable text (1) for partially sighted users (1) / full keyboard navigation (1) for users with motor impairments who cannot use a mouse (1) / plain language and consistent layout (1) for users with cognitive impairments or low literacy (1) / labelled form fields (1) for screen-reader users (1).

5 Distinguish between a static web page and a web application. [2]

Static page: stored as a finished file and simply retrieved and displayed, with no input or processing (1) / Web application: a program run through a browser that processes user input, often using a database, e.g. online banking (1).

6 A member of a credit union signs in to its online banking by entering a member number and password. Using this example, explain which processing happens on the client and which happens on the server. [4]

Client-side: the browser checks that both member number and password have been entered (1) / and that the member number has the correct format/length, prompting the user again without contacting the server (1) / Server-side: the server checks the member number and password against the stored records in the database (1) / then retrieves the account details, builds the account page and returns it to the browser to display (1).

7 A GAA club's membership form checks in the browser that the date of birth is filled in and in the correct format. Explain why the same data should also be validated on the server. [3]

Client-side checks can be bypassed, e.g. by disabling JavaScript, editing the page or sending a request directly (1) / so invalid or malicious data could reach the server (1) / server-side validation protects the integrity and security of the club's database (1).

8 Describe two tests that should be carried out on a website before it is published, giving the expected outcome of each. [4]

Any two tests, each with a description (1) and an expected outcome (1), e.g.: click every link (1) – each opens the correct page with no 404 errors (1) / submit the form with invalid data such as a blank email (1) – submission blocked and a clear error message shown next to the field (1) / view in different browsers and on a phone (1) – layout displays correctly with no sideways scrolling (1) / navigate using the keyboard only (1) – every link and control can be reached and used (1).

9 Describe the steps a small business in Derry~Londonderry would need to take to publish its new website on the internet. [4]

Register a domain name with a registrar (1) / choose and pay for a web hosting service (1) / upload the files to the host's server, e.g. by FTP/SFTP or a control panel (1) / point the domain's DNS records to the host (1) / install an SSL/TLS certificate so the site uses HTTPS (1) / test the live site (1). Max 4.

10 A leisure centre in Newry is replacing telephone bookings with an online booking web application used by people of all ages. Discuss the factors the developers should consider when designing, building and testing the application, including usability, accessibility and the division of processing between client and server. [8]

Indicative content: Usability – consistent menus, few steps to book, large buttons and readable fonts for older users, clear confirmation, responsive design for phones, quick loading, clear error messages; possibly a linear booking process. Accessibility – WCAG 2.2 principles (perceivable, operable, understandable, robust); alt text, contrast, keyboard access, labels, plain language; supporting users who previously phoned; legal duties. Client-side – presence, format and range checks for instant feedback and fewer wasted requests; showing price totals. Server-side – re-validate all input as client checks can be bypassed; check availability against the database, update places, record booking, take payment, roll back on failure, generate confirmation; protects personal data.

> Testing – test plan with valid/invalid/extreme data, broken links, cross-browser/device, accessibility and usability testing with real customers of different ages, performance at busy times. Levels: 7–8 wide range of well-developed points across all areas applied to the leisure centre, with a justified conclusion; 4–6 several relevant points with some explanation and context, may be unbalanced; 1–3 limited, mostly descriptive points with little application.

E Quick quiz

Circle one answer · 9 marks

1 An online shop checks in the browser that a card number is 16 digits before sending it. What kind of processing is this? [1]

A Server-side processing
B Client-side processing ✓
C Database processing
D Batch processing

2 Why must a web server re-validate data even if the browser has already checked it? [1]

A Browsers cannot perform any validation
B It makes the page load faster
C Client-side checks can be bypassed, so the server must protect its data ✓
D The server needs to change the page colours

3 Which navigation structure suits a step-by-step tutorial? [1]

A Linear ✓
B Hierarchical
C Mesh
D Random

4 Which WCAG principle covers being able to use a site with a keyboard only? [1]

A Perceivable
B Robust
C Understandable
D Operable ✓

5 Which task can ONLY be carried out server-side? [1]

A Checking a field is not empty
B Checking that a date is in the format dd/mm/yyyy
C Checking how many seats are still available on a flight ✓
D Showing a running total as items are ticked

6 What is a broken link? [1]

A A link to an external site
B A link that leads to a missing page ✓
C A link with no underline
D A link inside an image

7 What does a web host provide? [1]

A Server space and connectivity to store and serve the site's files ✓
B A domain name only
C A web browser
D An HTML editor

8 What does a storyboard show when planning a website? [1]

A The database tables
B The amount of traffic expected
C The cost of hosting
D The layout of each page, such as where menus, text and images appear ✓

9 What does responsive design do? [1]

A Speeds up the web server
B Encrypts form data
C Adapts the page layout to the size of the screen ✓
D Translates the page into other languages